The limits of magic-state cultivation

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Magic-state cultivation is a leading way to prepare the T states that fault-tolerant quantum computers need. Its circuits are designed with efficient simulations that replace every T gate by an S gate. This hides a loss of fault tolerance. An exact criterion reduces any cultivation circuit to one stabilizer code on its T gates and shows that current checks tolerate only half the faults their simulations promise. We identify the cause and design checks that avoid it. On a square grid, they keep the full fault distance and fail orders of magnitude less often than current checks.

INTRODUCTION

A fault-tolerant quantum computer needs a steady supply of T states, $|T\rangle = (|0\rangle + e^{i\pi/4}|1\rangle)/\sqrt{2}$, because the Clifford gates that error-correcting codes implement easily are not universal on their own. Magic-state cultivation [1] is one of the most economical ways known to make them. It prepares a T state in a small color code and checks it repeatedly with physical T gates, keeping the state only if every check passes. Simulations predict that cultivation reaches logical error rates that previously required distillation, at a cost comparable to that of a logical CNOT [1], and it has been demonstrated on hardware [2]. Variants cultivate on surface codes [3, 4] and prepare multi-qubit magic states [5, 6], and earlier schemes also inject a state and verify it by postselection [7–10].

These circuits have been designed with simulations that contain no T gates. Clifford circuits can be simulated efficiently and general circuits with T gates cannot, so cultivation circuits are simulated with every T gate replaced by an S gate, its Clifford square. We call this the S proxy. The proxy keeps the structure of every check, and the circuits that are proposed and built are chosen with it. Exact simulators that keep the T gates have appeared only recently [11–15], and they show that the proxy is not faithful: the distance-three and distance-five circuits of Ref. [1] tolerate fewer faults than their proxies [15, 16]. Simulators judge one circuit at a time. Three questions remain. Is there a criterion that decides fault tolerance with real T gates for every circuit and explains its loss? How much fault tolerance do current checks lose, and does the loss grow with the code distance? Can a check keep all of it?

We answer them in turn (Fig. 1). First, the fault tolerance of any cultivation circuit is decided by one stabilizer code on its T gates, together with a linear map that carries faults to Pauli errors on those T gates. Because the checks of cultivation are deterministic, this criterion has a closed form that can be evaluated in polynomial time. It also shows that cultivation needs at least as many T gates as distillation needs noisy inputs, and exactly as many if it ends in an ideal decoder. Second, current checks lose about half their distance. With real T gates, the GHZ-fold checks in use today tolerate only $(\delta-1)/2$ faults on color codes of distance $\delta = 3$, 5 and 7, while their proxies tolerate $\delta - 1$ at $\delta = 3$ and 5 and at least four at $\delta = 7$. A single mechanism is responsible. Third, checks that avoid this mechanism keep the full distance at $\delta = 3$ and 5, and at least six of seven at $\delta = 7$. They fit on a square grid with nearest-neighbor gates, and they fail orders of magnitude less often than the checks in use today.

CULTIVATION AND FAULT DISTANCE

A cultivation circuit injects a T state into a triangular color code and checks the logical state, possibly growing the code between checks, up to a final distance δ. Every nontrivial logical operator of the final code acts on at least δ physical qubits. A check applies a T or $\mathrm{T}^\dagger$ gate to every data qubit, measures the logical X operator and undoes these gates. The pattern of T and $\mathrm{T}^\dagger$ gates is chosen so that the check measures the logical operator $TXT^\dagger$, of which $|T\rangle$ is an eigenstate. A correct state therefore passes every check, and a state that fails is discarded. At the end an ideal decoder measures all stabilizers, keeps the state only if none is violated, and unencodes the logical qubit. This last step stands in for the later stages of a real protocol, in which the code is grown and protected by error correction, as in the simulations of Ref. [1]. Ref. [17] analyzes what these stages cost when the final postselection must use noisy syndrome data. Figure 1(a) sketches such a circuit, and Fig. S1(a) shows on the Bloch sphere what a single check does to an error.

Every operation of the circuit is noisy. A fault is a Pauli error after a preparation or a gate, or a wrong measurement outcome, and each location fails independently with probability of order p. A fault configuration is *harmless* if the circuit then rejects the state or outputs the correct one, and *harmful* if it can accept a state that differs from the correct one. The *fault distance* d is the smallest number of faults that can be harmful, so that

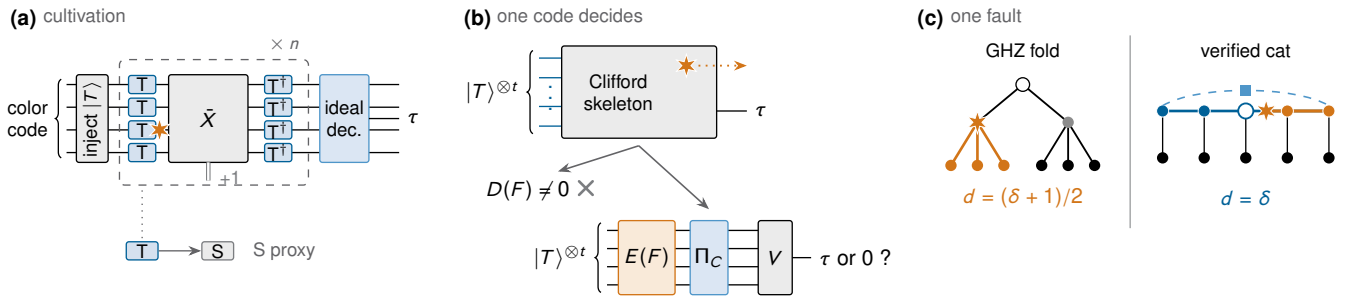


FIG. 1. The main ideas of this paper. (a) A cultivation circuit injects a T state into a color code and checks it n times. A check applies T gates to the data qubits, measures the logical X operator, keeps the outcome $+1$ and undoes the T gates, which together measures the logical operator $TXT^\dagger$. On larger codes the layer mixes T and $\mathrm{T}^\dagger$ gates. An ideal decoder ends the circuit. Faults (stars) can occur at every operation. The S proxy replaces every T gate by an S gate, which makes the circuit Clifford but changes how X errors behave. (b) Replacing every T gate by an injected T state leaves a noisy Clifford circuit, the skeleton, acting on t ideal T states. A fault pushed to the end of the skeleton (dotted) either fires a detector, $D(F) \neq 0$, and is rejected, or acts as a Pauli error $E(F)$ on the T inputs. It is harmless exactly when the code C still projects the perturbed inputs onto the target τ or onto zero (Theorem 1). (c) In a GHZ fold, one fault spreads X over a branch of data qubits (vermilion). Real T gates turn this into a coherent error that one more fault completes (Lemma 4), and the check keeps only fault distance $d = (\delta+1)/2$ on a code of distance δ. In a verified cat grown along a path from its middle (open circle), one fault spreads X only towards one end of the path. The parity of the two ends (dashed, measured through the square) catches it, and the check keeps the full fault distance $d = \delta$.

an accepted state is wrong with probability of order p^d. It cannot exceed the code distance δ, because δ errors on the data just before the decoder can form a logical error that the decoder does not see (Sec. S1). A circuit with $d = \delta$ keeps its full distance.

ONE CODE DECIDES

The obstacle to an exact analysis is that a Pauli fault before a T gate becomes a non-Pauli error after it. We therefore move the T gates out of the way. A T gate on a qubit can be replaced by a small Clifford circuit that consumes an ancilla prepared in $|T\rangle$: a CNOT from the qubit to the ancilla, followed by a projection of the ancilla onto $|0\rangle$ [18],

$$\sqrt{2}\,\langle 0|_a \,\mathrm{CX}_{q\to a}\,|\psi\rangle_q|T\rangle_a = T|\psi\rangle_q. \tag{1}$$

Doing this for all t T gates leaves a noisy Clifford circuit, the *skeleton*, that acts on t ideal T states. The skeleton can be followed exactly with the stabilizer formalism. Every fault can be pushed to its end, where it either contradicts a measurement outcome that is fixed in the skeleton, which we call a *detector*, or acts as a Pauli error on the t T states.

Theorem 1. *Consider a cultivation circuit, ending in its ideal decoder, with t T gates, without feedforward and with every measurement postselected, that prepares a target state τ when no fault occurs. Suppose that no Pauli operator other than the identity fixes τ, as is the case for a T state. Then there are a stabilizer code C on t qubits and two linear maps, D from fault configurations to detector patterns and E from fault configurations to Pauli operators on t qubits, such that a fault configuration F is harmless if and only if $D(F) \neq 0$ or*

$$\Pi_C\, E(F)\,|T\rangle^{\otimes t} \in \mathbb{C}\,\Pi_C|T\rangle^{\otimes t}, \tag{2}$$

where Π_C is the projection onto C. The map E is defined up to phases and stabilizers of C.

In words, a fault configuration is harmless exactly when it is detected or when the code C still projects the perturbed T states onto the correct encoded state, or onto zero [Fig. 1(b)]. The code C is fixed by the circuit and does not depend on the faults, which only choose the Pauli operator $E(F)$ that perturbs its inputs. Equation (2) is the condition that decides magic-state distillation, where perfect Clifford circuits measure a code on noisy T states [19]. Cultivation differs only through the maps D and E, which reject faults or carry them from anywhere in the circuit onto the T states. The pair (D, E) plays the role that the detector error model plays for Clifford circuits [20]. Exact simulators instead follow each fault through the T gates [11–15], and Refs. [16, 21] evaluate the Clifford error that a fault leaves at the end of the circuit. Here a single code, fixed by the circuit, judges every fault. Protocols with feedforward and random choices reduce to the circuit of a single accepted run without loss of fault distance (Proposition 7). The proofs are in Sec. S1.

An exact test. Condition (2) involves the state $|T\rangle^{\otimes t}$, whose dimension grows exponentially with t. In cultivation it can be decided without writing this state down, because every check measures an operator of which the target is an eigenstate. A Z error on the T inputs therefore either leaves the encoded state unchanged, flips the

logical qubit, or is rejected with certainty. An X error is different, because $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$ is a superposition of a correct and a Z-flipped T state. A Pauli error whose X part acts on $|a|$ inputs therefore spreads over $2^{|a|}$ Z patterns, and the code sorts each of them into the correct state, the wrong state or rejection. Adding up the amplitudes gives the following result, which is stated precisely and proved in Sec. S2.

Theorem 2. *If the checks of the circuit are deterministic in this sense (Definition 10), then for every fault configuration F with $D(F) = 0$, whose error $E(F)$ has an X part on $|a|$ inputs, there are two numbers $g_0(F)$ and $g_1(F)$, each a sum $\sum_z i^{q(z)}$ over a quadratic form q with values modulo four, such that F is accepted with probability $2^{-|a|}(|g_0|^2 + |g_1|^2)$ relative to the fault-free circuit and outputs a wrong state with probability $2^{-|a|}|g_1|^2$. In particular, F is harmful if and only if $g_1(F) \neq 0$, which is decided with $O(t^3)$ operations.*

Such Gauss sums are evaluated by eliminating one variable at a time. Each step either multiplies the sum by a simple factor or shows that it vanishes. Every fault configuration of a cultivation circuit can thus be judged exactly, with real T gates, at the cost of linear algebra. We combine this with a stim detector error model of the skeleton, which lists all fault classes with their detector patterns [20], and with a search for sets of faults whose detector patterns cancel. Fault distances are then certified with the same search as for Clifford circuits. A second, independent implementation follows the skeleton symbolically and decides Eq. (2) with state vectors. The two agree fault by fault on the distance-three circuit of Ref. [1] (Secs. S7 and S8).

The cost in T gates. Theorem 1 also bounds the number of T gates. Z faults on the T gates are never detected, and they act on the code C exactly like noise on the inputs of a distillation protocol. A circuit of fault distance d therefore contains a distillation protocol of order d. Conversely, every such protocol can be turned into a cultivation circuit of at least the same fault distance that ends in an ideal decoder of the protocol's code, as in the simulations of Ref. [1] (Sec. S3).

Theorem 3. *Every cultivation circuit of fault distance d for one T state contains at least as many T gates as the least number of noisy inputs of a distillation protocol of order d. This number of T gates suffices if the circuit ends in an ideal decoder for its output code, and it is at least $3d - 4$ for odd d and $3d - 6$ for even d.*

By Proposition 7, the lower bound holds for every accepted run of an adaptive protocol as well. Since distillation of order two needs nine noisy T states [19], fault distance two needs exactly nine T gates, and a nine-T circuit attains it (Sec. S5). The ideal decoder matters only for the converse. A noisy final stage is a Clifford circuit and adds no T gates, but the fault distance then also depends on how that stage is built. To our knowledge, the lower bounds are new for distillation as well, because the code of every order-d protocol has distance at least d (Lemma 8). With a linear-programming bound for stabilizer codes, they improve Ref. [19] for one T state from $d = 6$ on (Sec. S3). Published circuits are far from these bounds. The distance-five circuit of Ref. [1] uses 53 T gates for fault distance three, which fifteen T gates already reach, although our fifteen-T circuit accepts only one fault-free run in 1024 (Sec. S5). At practical distances, cultivation is limited by its checks, not by its number of T gates.

REAL T GATES HALVE THE DISTANCE

All the smallest harmful fault configurations that we found share one structure, which the following lemma describes. Consider a color code of distance δ and a set ℓ of δ data qubits that supports a logical X operator.

Lemma 4 (Coherent completion)**.** *Suppose that undetected faults leave an X error on a nonempty proper subset u of ℓ just before the last layer of T gates, that further faults put X on the rest of ℓ after this layer, and that the circuit only measures stabilizers between this layer and the ideal decoder. Then the decoder accepts with probability $2^{-|u|}$ relative to the fault-free circuit and outputs a state of fidelity $1/2$ with the target. With S gates in place of T gates, the same faults are detected.*

The reason is that a T gate turns an X error into a superposition, $TXT^\dagger \propto (I - iZ)X/\sqrt{2}$ [Fig. S1(a)]. After the last T layer, the error on u is a superposition of X^u with every Z error inside u. The decoder rejects every component that contains a Z error, since such an error acts on fewer than δ qubits of a minimum-weight logical support, so that it is neither a stabilizer nor a logical operator. It keeps the one component without a Z error, and the remaining faults complete X^u to a logical operator. An S gate instead turns X into Y, whose Z part the decoder always detects. The proof is in Sec. S1. It follows that a check keeps its full distance only if no s undetected faults can put X on more than s qubits of such a logical support before the last T layer (Sec. S10).

Current checks violate this condition. They measure the logical operator with a GHZ fold, a tree of CNOTs that collects the parity of the data onto one qubit, which is measured before the tree is undone. When the tree is undone, a single X fault on one of its qubits spreads to a whole branch of data qubits [Fig. 1(c)]. With S gates the decoder detects this spread. With T gates one more fault completes it coherently (Fig. 2). We certified this loss on the published circuits and on fold checks that we built at larger distance.

Proposition 5 (computer-assisted)**.** *With real T gates in place of the S gates of their simulations, (i) the*

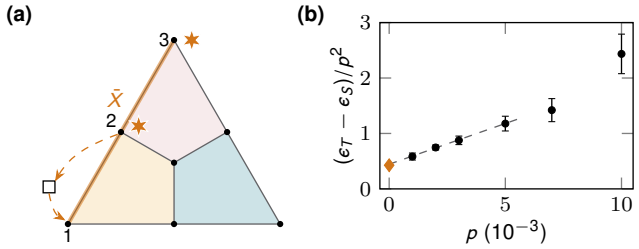


FIG. 2. How real T gates lose distance in the distance-three circuit of Ref. [1]. (a) One of its 48 harmful pairs of faults (stars) on the seven data qubits of the color code (faces shaded), following Ref. [16]. The first fault, an X error on qubit 2, is copied to qubit 1 through a qubit of the fold (square). In the last layer, S gates turn X_1X_2 into Y_1Y_2, which the top face detects. T gates turn it into $\frac{1}{2}(X+Y)_1(X+Y)_2$, whose component X_1X_2 passes every check, and the second fault, on qubit 3, completes the logical operator $\bar{X}$ along the left edge. (b) Difference of the logical errors per accepted run with T and with S gates, divided by p^2, in the state-vector simulations published with the circuit [22] (dots, one standard deviation), with a weighted linear fit as a consistency check (dashed). The diamond is the exact limit 32/75 at $p \to 0$ from Theorem 1.

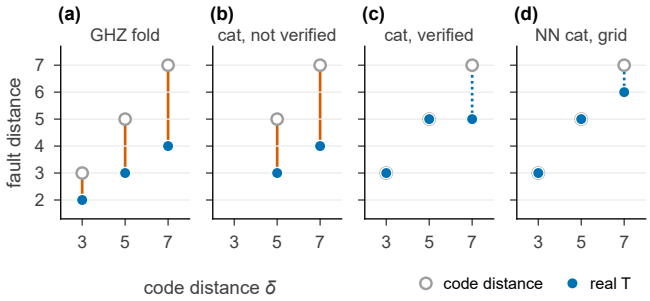


FIG. 3. Fault distance with real T gates (filled) and code distance δ (open), which the S proxy reaches. The vermilion gaps are the distance lost to real T gates. (a) GHZ folds, including the circuits of Ref. [1]. (b), (c) Cat-state checks without and with verification. (d) Nearest-neighbor cat checks on a square grid. At $\delta = 7$ the verified cats are certified only from below, at five in (c) and six in (d), and the dotted ranges are open.

distance-three and distance-five cultivation circuits of Ref. [1] have fault distance two and three, while their proxies have three and five, and (ii) GHZ-fold checks of the same kind on color codes of distance three, five and seven have fault distance two, three and four, for every fold tree we tried.

In every case the fault distance with real T gates is $(\delta + 1)/2$, about half the code distance, and the gap grows with δ (Fig. 3). These failures are partly suppressed, since by Lemma 4 they are accepted only with probability $2^{-|u|}$. Even so, the leading term of the logical error of the distance-three circuit of Ref. [1] is $32p^2/75$, in agreement with the exact series of Ref. [15]. Because this term is of order p^2 instead of p^3, the proxy underestimates the logical error by a factor that grows as p decreases. The circuit run on hardware in Ref. [2] also has fault distance two with real T gates, against three for its proxy (Sec. S12). Secs. S9 and S10 give the details, including flags that partly repair the fold.

KEEPING THE DISTANCE

Lemma 4 points to the remedy: no single fault should put X on more than one data qubit before the last T layer. A Shor-style check [23] has this property. It prepares a cat state $(|0\cdots0\rangle + |1\cdots1\rangle)/\sqrt{2}$ on fresh qubits, verifies it by measuring parities of its qubits before it touches the data, couples each cat qubit to one data qubit by a CNOT, and reads the outcome of the check from the parity of the cat qubits in the X basis. A fault in the preparation that spreads X over several cat qubits is caught by the verification, and every other fault reaches at most one data qubit.

Such a check needs no long-range gates. On a square grid we grow the cat along a path of qubits from its middle outward, one CNOT per step in each direction. A single X fault on a path qubit then spreads only towards the end of the path beyond it, and the parity of the two ends, measured through an ancilla next to both, catches every such spread [Fig. 1(c)]. Each data qubit sits next to its own path qubit. After the coupling we undo the growth, so that the outcome of the check sits on one qubit and the other path qubits return to $|0\rangle$, where measuring them gives further detectors. At $\delta = 3$ the path runs around a rectangle, which the ancilla closes. At $\delta = 5$ the path folds into a U on three rows, and the middle row measures the parities across the U and along its diagonals, which together catch every spread caused by two faults (Fig. 4).

Proposition 6 (computer-assisted)**.** *Checks with a verified cat keep the full fault distance with real T gates: five on the distance-five color code and at least five on the distance-seven code with all-to-all connectivity, where the same checks without verification have three and four. With nearest-neighbor gates on a square grid, the ring check on the distance-three code (17 qubits) and the ladder check on the distance-five code (46 qubits) have fault distance three and five, and the ladder on the distance-seven code (91 qubits) has at least six.*

These checks are also cheap. We compare them under SI1000 noise, a standard model of superconducting hardware in which two-qubit gates fail with probability p, measurements with $5p$ and single-qubit gates with $p/10$ [24]. At $p = 10^{-3}$ the ring accepts 88% of runs and fails with probability $0.056\,p^3 + 30\,p^4 \approx 9 \times 10^{-11}$. The best distance-three GHZ fold that we built fails about 600 times more often ($0.049\,p^2$), accepts 43% of runs and

TABLE I. Fault distance with S gates (proxy) and with real T gates, certified exactly, and the leading term of the probability of an accepted wrong output under uniform noise of strength p, for checks on color codes of distance δ. The first two rows are the circuits of Ref. [1]. The other checks start from an ideally encoded state. For the GHZ fold at $\delta = 7$, the leading term ranges over the fold trees we tried.

check	δ	proxy	real T	leading term
GHZ fold [1]	3	3	2	$0.43\,p^2$
GHZ fold [1]	5	5	3	$1.5\,p^3$
GHZ fold	7	≥ 5	4	$(0.011 \text{ to } 15)\,p^4$
cat, not verified	5	5	3	$0.54\,p^3$
cat, not verified	7	≥ 5	4	$0.093\,p^4$
cat, verified	5	5	**5**	$3.5 \times 10^3 p^5$
cat, verified	7	≥ 5	$\geq$ **5**	
grid, ring	3	3	**3**	$3.6\,p^3$
grid, ladder	5	5	**5**	$12\,p^5$
grid, ladder	7	≥ 5	$\geq$ **6**	

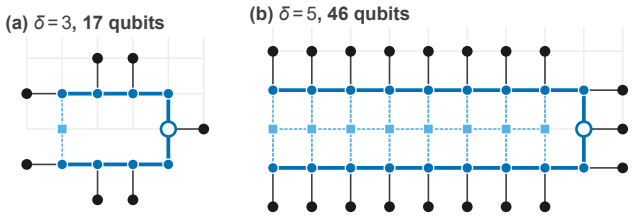


FIG. 4. Nearest-neighbor cat checks on a square grid: data qubits (black), the cat path (blue) grown from its middle (open circle), and verification ancillas (light blue) with the parities they measure (dashed).

needs more than twice the qubits. The ladder accepts 39% of runs, against 2% to 9% for our folds and unverified cats of the same distance, mostly because it needs no stabilizer measurements between checks. Without undoing the growth it accepts 55% of runs and fails with probability 4×10^{-14}. The published distance-five protocol, which also prepares its state, reaches 2×10^{-9} at 16% acceptance. Table I collects the fault distances and Sec. S11 the costs.

These checks start from an ideally encoded state, like those of Proposition 5(ii). At distance three we also built a complete nearest-neighbor protocol in which every operation is noisy. It keeps fault distance three, so preparation does not limit the fault distance there. Its cost lies in the stabilizer measurements around the checks, which our version performs slowly, and it beats the circuit of the experiment of Ref. [2] only for $p \lesssim 3 \times 10^{-4}$ (Sec. S11).

OUTLOOK

Cultivation has so far been judged by simulations that replace its T gates by S gates. Theorem 1 replaces this substitution by an exact criterion, a single stabilizer code on the T gates, and Theorem 2 makes the criterion fast enough to certify realistic circuits. With it we find that the checks in use today keep only about half their distance, for the reason that Lemma 4 identifies, and that checks built to avoid it keep all of it, with nearest-neighbor gates and a far lower logical error. Our results assume stochastic Pauli noise, as the simulations of cultivation do. Coherent errors and leakage, which experiments also see [2] and which can raise the cost of cultivation [25], are outside this model. The main open problems are a cheap way to measure stabilizers on the grid around these checks, a certificate that the ladder keeps the full distance seven, and lower bounds on the gates that real-T checks need (Sec. S13).

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Data availability. The scripts, circuits and results are available at `https://github.com/llorebaga/Bounds_magic`. The exact evaluation of Theorem 2 and the circuits of Propositions 5 and 6 are in its directory `cultivation_exact`. The circuit chunks and state-vector data of Ref. [22] are reproduced there under the Apache License 2.0, and the circuits of Ref. [2] are decoded from the links published with that paper.

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Supplemental Material

S1. THE CRITERION

This section states the setting precisely and proves Theorem 1, Lemma 4 and the bound of the fault distance by the code distance. We first fix the circuits, the faults and the decoders that the main text describes informally.

Circuits and protocols. Throughout, $|T\rangle = (|0\rangle + \omega|1\rangle)/\sqrt{2}$ with $\omega = e^{i\pi/4}$. We begin with the most general protocols we consider. A *Clifford+T protocol* acts on qubits initialized in $|0\rangle$. It may use Clifford gates, T and $\mathrm{T}^\dagger$ gates, Pauli measurements, classical randomness, feedforward of any operation on earlier outcomes, postselection and discarding, and it outputs r qubits. Its target is a pure state τ with stabilizer nullity $\nu(\tau) = r$. This means that τ is fixed by no Pauli operator other than the identity, as is the case for $|T\rangle$ and $|\mathrm{CCZ}\rangle$. The circuits of the main text belong to a simpler class. A protocol is *non-adaptive and fully postselected* if it is a fixed sequence of Clifford gates, T and $\mathrm{T}^\dagger$ gates, projections $(I + sQ)/2$ with a Pauli Q and a fixed sign s, and removals $\langle k|_q$ of qubits with fixed $k \in \{0, 1\}$. We count $\mathrm{T}^\dagger$ gates as T gates. Such a circuit has a single Kraus operator K, the linear map that it applies to its input. We call the circuit *correct* if $K|0\rangle$ is a nonzero multiple of $|\tau\rangle$.

Faults. Next we fix the noise model. Every preparation, gate and measurement is a *location*, except those declared ideal. Removals are ideal. A fault at a location is either a Pauli operator on its support or the flip of a projection's sign. A Pauli fault acts immediately after a preparation or gate, or immediately before a projection. Faults at different locations compose. The *fault space* $\mathbb{V}$ is the $\mathbb{F}_2$-vector space of all assignments of Pauli operators and flips to locations. The weight $\mathrm{wt}(F)$ of $F \in \mathbb{V}$ is the least number of single faults that compose to F. It is the number of locations at which F acts, except that a Pauli fault and a flip at the same measurement count twice. Each location suffers each of its faults independently with probability $p\lambda > 0$, with a constant λ for each kind of fault. A run of an adaptive protocol is therefore described by its random choices and observed outcomes b, together with the faults F at the locations executed along b, and it has a Kraus operator $K_{b,F}$. The pair (b, F) is *harmless* if $K_{b,F}|0\rangle \in \mathbb{C}|\tau\rangle$, which includes rejection. The output error after postselection is $\epsilon = O(p^d)$ exactly when every run with $\mathrm{wt}(F) < d$ is harmless and some fault-free run is accepted. The largest such d is the *fault distance*. It depends only on which faults are possible, not on their probabilities, as long as these are of order p. It is therefore the same, for example, under depolarizing noise, which applies one random Pauli operator per location. We call the harmful fault configurations of the least weight d *malignant*. Together they give the leading term of the output error.

Ideal decoders. The output of cultivation is encoded, so we must also say how an encoded output is judged. A protocol whose output is encoded is judged after an ideal decoder. This decoder consists of projections onto the stabilizers of an $[[m, r, \delta]]$ code Q on m output qubits, followed by an ideal Clifford unencoding. We call the decoder *admissible* if the fault-free state at its input lies in Q, so that it accepts the fault-free run with certainty. Its projections and the unencoding are ideal operations of the circuit, so the results below apply to the circuit together with its decoder. Simulations of cultivation use decoders of this kind [1]. The admissibility condition is needed. Without it, the decoder could carry out the T-dependent checks itself. For example, nine noisy T states handed to an ideal decoder of the code of Proposition 16 below would already tolerate every single fault.

Reduction to one run. A protocol may branch on its random choices and outcomes. For the fault distance it suffices to study one accepted run, as the following proposition shows.

Proposition 7. *Let a Clifford+T protocol have fault distance d, and let b be any accepted fault-free run. The non-adaptive, fully postselected circuit formed by the operations executed along b, with every measurement replaced by the projection onto its outcome in b and every discarded qubit removed with an outcome of nonzero amplitude, is correct and has fault distance at least d.*

The proof below uses only one fact: a sum of positive operators is a multiple of a pure state exactly when each term is. As a consequence, every lower bound on the qubits, gates or T gates of correct non-adaptive circuits of fault distance d holds for every accepted fault-free run of every protocol of fault distance d.

The skeleton. In this notation, Theorem 1 states in addition that C has dimension 2^r with $\nu(\tau) = r$, that $\Pi_C|T\rangle^{\otimes t} \neq 0$, that the fault-free skeleton acts as $c\,V\Pi_C$ with a Clifford isometry V that maps $\Pi_C|T\rangle^{\otimes t}$ to a multiple of $|\tau\rangle$, and that a Z fault immediately before or after the jth T gate has $D = 0$ and $E = Z_j$. The proof is given below. The pair (D, E) plays the role of the detector error model of a Clifford circuit [20, 26, 27], with the logical observables replaced by the Pauli group on the T gates and the test on them by the collapse condition (2). Both maps are computed in polynomial time by following symbolically the Choi state of the skeleton, the state that it produces from one half of t Bell pairs. Only the collapse condition involves the T states (Sec. S7).

Detectors and checks. A measurement whose outcome is fixed in the circuit need not be a detector of the skeleton, and this difference matters for T gates. The entries of D belong to the projections whose outcome is fixed in the skeleton. Some projections have an outcome that is fixed in the circuit but depends on the T inputs in the skeleton. A check after a layer of T gates is an example. Such a projection is not a detector in this sense. It is part of the code C, and it rejects a fault only through Eq. (2). That is, it removes only the components of $E(F)|T\rangle^{\otimes t}$ that leave the code. In the distance-three circuit of Ref. [1], 4 of the 20 projections are of this kind: the X measurement that reads out the logical check, and three X-type stabilizers measured after the final T layer. Wan and Zapirain likewise find that 16 of its 20 detectors remain Pauli webs with T gates [15]. With S gates the circuit is Clifford, and all 20 are detectors.

Related descriptions. Several earlier works describe how faults act in cultivation circuits, and we relate our description to them. Replacing T gates by gadgets is standard in simulation [28]. For cultivation, Surti *et al.* observed that every Pauli fault reaches the end of the circuit as a Clifford error [21]. Chan *et al.* computed the acceptance and fidelity of a code state under such an error [16]. Exact simulators follow faults through the T gates [11, 12, 15, 29–32]. Theorem 1 instead moves the non-Clifford part to the inputs. There one code, independent of the fault, judges a Pauli operator that depends linearly on the fault.

X errors on the inputs. The new feature relative to distillation is that $E(F)$ may have an X part. On one qubit, $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$ and $XZ|T\rangle = -(i|T\rangle + Z|T\rangle)/\sqrt{2}$. A Pauli operator $E = X^u Z^v$ therefore turns $|T\rangle^{\otimes t}$ into a superposition of the states $Z^{v+w}|T\rangle^{\otimes t}$ with w supported on u, with amplitudes of modulus $2^{-|u|/2}$. To compare such operators with Z patterns, we use a weight that ignores phases and stabilizers. For a Pauli operator E on the T inputs, let $|E|_S$ be the least number of qubits on which an element of $E\langle iI, S_C\rangle$ acts.

Lemma 8. *(i) If every pattern $v + w$ with $\operatorname{supp} w \subseteq \operatorname{supp} u$ satisfies the distillation condition $\Pi_C Z^{v+w}|T\rangle^{\otimes t} \in \mathbb{C}\,\Pi_C|T\rangle^{\otimes t}$, then $X^u Z^v$ satisfies Eq. (2), and every F with $E(F) = X^u Z^v$ is harmless. (ii) If every Z pattern of weight below d satisfies the distillation condition, then every Pauli operator E with $|E|_S < d$ satisfies Eq. (2), and C has distance at least d.*

Part (i) says that an X error on the inputs is harmless when each of its Z components is. The smallest instance of (i) is a single check (Fig. S1(a)). A projection onto $|T\rangle$ rejects $Z|T\rangle$, and it returns $X|T\rangle$ and $Y|T\rangle$ to $|T\rangle$ half of the time. X and Y faults before the check are therefore harmless, although they are neither detected nor stabilizers (Sec. S6). The converse of (i) can fail, because harmful components may cancel. For the code on two qubits with the single stabilizer X_1X_2, the error $E = X_1X_2$ is a stabilizer and therefore harmless, while the Z patterns Z_1 and Z_2 in its expansion are harmful. Part (ii) is where cultivation and distillation meet. A code that tolerates Z faults on fewer than d inputs tolerates every Pauli fault on them. Since no Pauli operator fixes the target, such a code also has no logical operator of weight below d.

Proof of Proposition 7. We compare the accepted output of the whole protocol with the contribution of a single run. The unnormalized accepted output of the protocol is $\sum_{b',F'} q_{b',F'}\, K_{b',F'} \rho_0 K_{b',F'}^\dagger$. The sum runs over accepted runs b' and the fault configurations F' along them, with probabilities $q_{b',F'} > 0$. This expression is exact for stochastic Pauli noise, because faults are sampled classically at the locations actually executed. A run (b, F) is harmless exactly when its own term is a multiple of $|\tau\rangle\langle\tau|$. Now fix an accepted fault-free run b. Along b, every random choice and every measurement outcome is fixed, so the operations executed along b form a non-adaptive circuit. For every fault configuration F at its locations, $K_{b,F}$ is the Kraus operator of this circuit with F inserted, postselected on the outcomes of b. A discarded qubit contributes a partial trace, $\sum_k \langle k|\cdot|k\rangle$. If the sum is a multiple of a pure state, each term is. Replacing the partial trace by the removal $\langle k|$, for an outcome k with $\langle k|K_{b,\emptyset}|0\rangle \neq 0$, therefore keeps the circuit correct and every harmless configuration harmless. Hence every configuration with $\mathrm{wt}(F) < d$ is harmless in the circuit. The removals are ideal and add no locations. □

Proof of Theorem 1. The proof has five steps. We build the skeleton, show that the signs of its projections act as Pauli operators, move the faults onto the T inputs, derive the collapse condition, and finally treat Z faults at T gates.

Skeleton. Eq. (1) follows from the state $\mathrm{CX}_{q\to a}(\alpha|0\rangle + \beta|1\rangle)(|0\rangle + \omega|1\rangle)/\sqrt{2}$, whose component with a in $|0\rangle$ is $(\alpha|0\rangle + \beta\omega|1\rangle)/\sqrt{2}$. We introduce one ancilla a_j per T gate, prepared ideally in $|T\rangle$, and write $\mathrm{T}^\dagger = S^\dagger T$ with an ideal $S^\dagger$. A fault of the jth T gate acts after the gadget. The circuit becomes $K_F = 2^{t/2} K_{S,F}(|0\rangle \otimes |T\rangle^{\otimes t})$. Here the skeleton $K_{S,F}$ consists of Clifford gates, projections and removals, and F is inserted at the same qubits and times. Let $M_F = K_{S,F}(|0\rangle \otimes \cdot)$, a linear map from t qubits to r qubits, and let $M = M_0$. As in the proof of Theorem 1 of Ref. [19], M applied to half of t Bell pairs gives a stabilizer state on $t + r$ qubits. By the local-Clifford normal form of bipartite stabilizer states, $M = c\, V \Pi_C$ with a stabilizer code C on t qubits and a Clifford isometry V from C onto its range. The range contains $|\tau\rangle$ and is a stabilizer subspace. Since no Pauli operator other than the identity fixes τ, the range is the whole output space. So $\dim C = 2^r$, and V is unitary from C onto $(\mathbb{C}^2)^{\otimes r}$.

Signs. We treat each removal $\langle k|_q$ as a projection of Z_q with sign $(-1)^k$ followed by an ideal removal. Let $\sigma \in \mathbb{F}_2^\mu$ list the signs of all μ projections relative to the circuit, so that $M^{(0)} = M$. Let $|\Psi^{(\sigma)}\rangle = (M^{(\sigma)} \otimes I) \sum_x |x\rangle|x\rangle$ be the unnormalized Choi vector. If we follow the circuit with the stabilizer formalism [33], the Choi vector stays a pure stabilizer state. The Pauli of every projection therefore either anticommutes with a current stabilizer or is one up to sign. In the first case the projection halves the squared norm and replaces that stabilizer by the projected Pauli with the chosen sign. In the second case it keeps the state or annihilates it, depending on the sign. The unsigned stabilizer groups and the norms of the nonzero $|\Psi^{(\sigma)}\rangle$ are therefore independent of σ. The signs required by the deterministic projections are affine functions of the earlier signs, so $|\Psi^{(\sigma)}\rangle \neq 0$ exactly when $D_0\sigma = 0$ for a linear map D_0. Two stabilizer states with the same unsigned group and norm differ by a Pauli operator and a phase. This Pauli operator is determined by the sign difference modulo the stabilizer group, and it depends on it linearly. Hence, for $D_0\sigma = 0$, $|\Psi^{(\sigma)}\rangle = \zeta_\sigma(A_\sigma \otimes B_\sigma)|\Psi^{(0)}\rangle$ with $|\zeta_\sigma| = 1$ and Pauli operators A_σ on the output and B_σ on the reference. In other words, $M^{(\sigma)} = \zeta_\sigma A_\sigma M B_\sigma^{\mathsf{T}}$.

Faults. We move a Pauli P inserted after an operation to the end of the circuit. A Clifford gate G turns it into $GPG^\dagger$. A projection $(I + sQ)$ with which it anticommutes becomes $(I - sQ)$ as P passes it. A removal absorbs the Z part of P on the removed qubit, and its projection turns the X part into a sign flip. A sign flip changes σ directly. Hence $M_F = \zeta\, P_{\mathrm{out}}(F)\, M^{(s(F))}$ with a phase ζ, a Pauli $P_{\mathrm{out}}(F)$ on the output and a sign vector $s(F)$, both linear in F. We set $D(F) = D_0 s(F)$, where D_0 has one row per projection, nonzero only for the projections whose outcome is fixed in the skeleton. If $D(F) \neq 0$, then $M_F = 0$ and the configuration is rejected. Otherwise $M_F = \zeta' P_{\mathrm{out}} A_s M B_s^{\mathsf{T}}$. It remains to move the output Pauli onto the inputs. For an output Pauli R, write $V\Pi_C = W(I \otimes \langle 0|^{\otimes(t-r)})U\Pi_C$, with a Clifford unitary U on t qubits that maps C onto $(\mathbb{C}^2)^{\otimes r} \otimes |0\rangle^{\otimes(t-r)}$ and a Clifford unitary W on r qubits. Then $RM = ML$ with the Pauli $L = U^\dagger(W^\dagger R W \otimes I)U$, which commutes with the stabilizers of C. Hence $M_F = \zeta'' M\, E(F)$ with $E(F) = L(P_{\mathrm{out}} A_s) B_s^{\mathsf{T}}$. Finally, $ME \propto ME'$ holds exactly when $E'E^\dagger$ acts on C as a scalar, that is, when it lies in $\langle iI, S_C\rangle$. This equivalence holds because V is injective on C, and because a Pauli that anticommutes with a stabilizer maps C to an orthogonal code. So E is well defined and linear modulo $\langle iI, S_C\rangle$ on the configurations with $D(F) = 0$, and we extend it linearly to all of $\mathbb{V}$.

Collapse. With these maps, $K_F|0\rangle = 2^{t/2}\zeta'' c\, V\Pi_C E(F)|T\rangle^{\otimes t}$ and $K_0|0\rangle \propto V\Pi_C|T\rangle^{\otimes t} \propto |\tau\rangle$. Since V is injective on C, $K_F|0\rangle \in \mathbb{C}|\tau\rangle$ exactly when Eq. (2) holds.

Z faults at T gates. A Z fault after the gadget of the jth T gate on qubit q satisfies $Z_q\langle 0|_a \mathrm{CX}_{q\to a} = \langle 0|_a Z_a Z_q \mathrm{CX}_{q\to a} = \langle 0|_a \mathrm{CX}_{q\to a} Z_a$. Here we used $\langle 0|_a = \langle 0|_a Z_a$ and $\mathrm{CX}_{q\to a} Z_a = Z_q Z_a \mathrm{CX}_{q\to a}$. The fault thus acts as Z on the ideal input a_j, so $D = 0$ and $E = Z_j$. A Z fault before the gate gives the same result, because Z commutes with T and with $S^\dagger$. □

Proof of Lemma 8. Part (i) expands a coherent error into its Z components, and part (ii) reduces a light Pauli operator to part (i). (i) On one qubit, $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$, $XZ|T\rangle = -(i|T\rangle + Z|T\rangle)/\sqrt{2}$ and $Y = iXZ$. Up to a phase, therefore, $X^u Z^v |T\rangle^{\otimes t} = \sum_w \gamma_w Z^{v+w}|T\rangle^{\otimes t}$, summed over w with $\operatorname{supp} w \subseteq \operatorname{supp} u$ and with $|\gamma_w| = 2^{-|u|/2}$. If each projected term lies on the line $\mathbb{C}\,\Pi_C|T\rangle^{\otimes t}$, so does their sum, and Theorem 1 applies. (ii) Multiplying E by an element of $\langle iI, S_C\rangle$ changes $\Pi_C E|T\rangle^{\otimes t}$ only by a phase. We may therefore take $E = X^u Z^v$ acting on fewer than d qubits. Every pattern $v + w$ in (i) is supported on these qubits, so it has weight below d, and (i) applies. For the distance, let E commute with S_C and have $|E|_S < d$. Then E commutes with Π_C, and $E\Pi_C|T\rangle^{\otimes t} = \Pi_C E|T\rangle^{\otimes t}$ is a multiple of $\Pi_C|T\rangle^{\otimes t}$. The multiple is nonzero because E is unitary. Through the isometry V, E acts on C as a Pauli operator P on r qubits up to a phase, and $|\tau\rangle$ is an eigenvector of P. A Hermitian multiple P' of P has eigenvalues ± 1, so P' or $-P'$ fixes $|\tau\rangle$. Since $\nu(\tau) = r$, $P' = \pm I$. Hence E acts on C as a scalar, that is, $E \in \langle iI, S_C\rangle$. Every logical operator outside $\langle iI, S_C\rangle$ therefore acts on at least d qubits. □

Proof of Lemma 4. We expand the error after the last T layer into Pauli components and follow each component through the decoder. For $U_q = T^{\pm 1}$, $U_q X U_q^\dagger = \bar{\omega}^{\pm 1} S^{\pm 1} X$ and $S^{\pm 1} = \omega^{\pm 1}(I \mp iZ)/\sqrt{2}$. Up to a phase, the error immediately after the layer is therefore $\prod_{q\in u}(I \mp iZ_q)X^u/2^{|u|/2} = 2^{-|u|/2}\sum_{w\subseteq u} c_w Z^w X^u$ with $c_w = \prod_{q\in w}(\mp i)$. The faults after the layer turn each component into $X^{\ell\setminus u} Z^w X^u = \pm Z^w X^\ell$. The stabilizer measurements and the ideal decoder project onto the reference syndrome. The operator X^ℓ is a logical operator, so it commutes with all stabilizers. The operator Z^w with $w \subseteq u \subsetneq \ell$ has $|w| < \delta$, so it is not a nontrivial logical operator. It therefore commutes with all X stabilizers only if it is a Z stabilizer. No Z stabilizer is supported inside ℓ: since the color code is self-dual, the X stabilizer on the same support would multiply X^ℓ to a lighter logical operator. Hence $w = \emptyset$. The projection maps every other component to an orthogonal syndrome space and removes it. The accepted component is $2^{-|u|/2}X^\ell$ applied to the fault-free state. It has relative probability $2^{-|u|}$, and it decodes to $X_L|\tau\rangle$. For $|\tau\rangle = |T\rangle$, $|\langle T|X|T\rangle|^2 = 1/2$, so the output has fidelity $1/2$. In the proxy, $SXS^\dagger = Y$ up to a phase, so the error after the layer

is the Pauli $Z^u X^\ell$ up to a phase. The operator Z^u with $\emptyset \neq u \subsetneq \ell$ is neither a stabilizer nor a logical operator. It has a nontrivial syndrome, and the configuration is rejected. □

The same expansion settles the two neighboring cases, which also occur among the malignant configurations of Sec. S9. First, if the completing faults put Y instead of X on one qubit q of $\ell \setminus u$, the accepted component must contain $Z^u Z_q$, which has a trivial syndrome only if $u \cup \{q\} = \ell$. For $|u| = \delta - 1$ the configuration is therefore accepted with probability $2^{-|u|}$ and outputs $Y_L|\tau\rangle$ up to a phase, again of fidelity $1/2$, and for smaller u it is rejected. Second, let the faults leave $X^\ell Z^y$ with $y \subseteq \ell$ before the last T layer, so that X covers all of ℓ. Then exactly the two components with $w = y$ and $w = \ell \setminus y$ survive, and the configuration is accepted with probability $2^{1-\delta}$. The decoded output is proportional to $X_L(I + cZ_L)|\tau\rangle$, where $c = \pm i$ because δ is odd, and c changes sign with the parity of $|y|$, since $c_{\ell\setminus y}/c_y = (-1)^{|y|} c_\ell$. For one sign the output is $|T\rangle$ and for the other it is orthogonal to $|T\rangle$. If the check is correct, $y = \emptyset$ is harmless: on the $[[7, 1, 3]]$ code, for every pattern of T and $\mathrm{T}^\dagger$ gates whose check fixes the encoded T state, $UX^\ell U^\dagger$ acts on the code as the check operator for every minimum-weight support ℓ. A Y error on one qubit of ℓ together with X on the rest therefore gives an output orthogonal to $|T\rangle$. We confirmed the lemma and both cases with state vectors on the $[[7, 1, 3]]$ code.

The code distance bounds the fault distance. Finally, the fault distance cannot exceed the distance of the code of an ideal decoder.

Proposition 9. *If a correct circuit ends with an admissible ideal decoder for an $[[m, r, \delta]]$ code, and each of the m qubits has an idle location immediately before the decoder, then its fault distance d is at most δ, and $m \geq r + 2(d-1)$.*

Proof. A logical Pauli of weight δ placed at the idle locations commutes with the stabilizers and passes the decoder. It then applies a nontrivial Pauli to the decoded state, and the result differs from $|\tau\rangle$ because $\nu(\tau) = r$. The inequality is the quantum Singleton bound [34]. □

S2. DETERMINISTIC CHECKS AND GAUSS SUMS

This section proves Theorem 2 of the main text in a more precise form. Lemma 8(i) is only a sufficient condition for harmlessness. When the checks are deterministic, in the following sense, the collapse condition can be decided exactly in polynomial time.

Definition 10. A correct circuit for $|T\rangle$ has *deterministic checks* if there are a subspace $\Sigma \subseteq \mathbb{F}_2^t$ and $\ell \in \mathbb{F}_2^t$ such that $Z^e \in \pm S_C$ for $e \in \Sigma$, Z^ℓ commutes with S_C and $VZ^\ell V^\dagger$ acts on the output as Z, and $\Pi_C Z^e |T\rangle^{\otimes t} = 0$ for every $e \notin \Sigma \cup (\ell + \Sigma)$.

Here Σ collects the Z patterns that are stabilizers of C up to sign, and Z^ℓ is the logical Z. Every Z error on the T gates is then a stabilizer of C, the logical Z, or rejected with certainty. This is the case when every check measures an operator of which the target is an eigenstate. It is not the case for code-based distillation, where Z errors on the inputs are detected only with some probability. For $e \in \Sigma$ we write $Z^e = \xi(e) g_e$ with $g_e \in S_C$. These signs define a character ξ of Σ, that is, a homomorphism to $\{\pm 1\}$, which we extend to a character χ of $\mathbb{F}_2^t$.

Theorem 11. *For a circuit with deterministic checks and a configuration with $D(F) = 0$ and $E(F) = X^a Z^b$,*

$$\Pi_C E(F) |T\rangle^{\otimes t} = \mu \, 2^{-|a|/2} \big(g_0 + g_1 Z^\ell\big) \Pi_C |T\rangle^{\otimes t}, \tag{S1}$$

with $|\mu| = 1$ and, for $c \in \{0, 1\}$,

$$g_c = \sum_{w \leq a,\ w+b \in c\ell + \Sigma} i^{|w|} \, \chi(w + b + c\ell). \tag{S2}$$

The configuration is accepted with probability $2^{-|a|}(|g_0|^2 + |g_1|^2)$ relative to the fault-free run, its wrong output has weight $2^{-|a|}|g_1|^2$, and it is harmful if and only if $g_1 \neq 0$.

In Eq. (S2), $w \leq a$ means that the support of w lies inside the support of a. We call the weight $2^{-|a|}|g_1|^2$ of the wrong output the bad weight. Theorem 2 of the main text restates this theorem, with a the X part of $E(F)$ and with Proposition 12 below for the cost.

Definition 10 removes every Z pattern outside the two cosets Σ and $\ell + \Sigma$, and the two surviving terms are orthogonal and of equal norm because $\langle T|Z|T\rangle = 0$. Each g_c is an exponential sum $\sum_{z \in \mathbb{F}_2^m} i^{q(z)}$ of a $\mathbb{Z}_4$-valued quadratic form, as the proof of the following proposition shows.

Proposition 12. *Each g_c is computed with $O(t^3)$ operations, and $|g_c|$ is zero or a power of $\sqrt{2}$. In particular the acceptance and the bad weight of every fault configuration are dyadic rationals.*

The coherence of real T gates enters only through these sums. Judging the terms of Eq. (S2) one at a time, and calling a configuration harmful when one of them lies in $\ell + \Sigma$, would give fault distance one for every circuit we studied: a single X fault before a check has such a term, which the check removes only through the cancellation $g_1 = 0$.

Proof of Theorem 11. We expand $X^a Z^b |T\rangle^{\otimes t}$ in Z patterns and apply Definition 10 to each term. On one qubit, $|T\rangle = (|0\rangle + \omega|1\rangle)/\sqrt{2}$, $Z|T\rangle = (|0\rangle - \omega|1\rangle)/\sqrt{2}$ and $X|T\rangle = \omega(|0\rangle + \bar{\omega}|1\rangle)/\sqrt{2}$. Writing $|0\rangle + \bar{\omega}|1\rangle = \alpha(|0\rangle + \omega|1\rangle) + \beta(|0\rangle - \omega|1\rangle)$ gives $\alpha + \beta = 1$ and $\alpha - \beta = \bar{\omega}^2 = -i$. Hence $\alpha = \bar{\omega}/\sqrt{2}$, $\beta = i\bar{\omega}/\sqrt{2}$ and $X|T\rangle = (|T\rangle + iZ|T\rangle)/\sqrt{2}$. Taking the tensor product over $\operatorname{supp} a$ and using $X^a Z^b = (-1)^{|a \wedge b|} Z^b X^a$, we obtain

$$X^a Z^b |T\rangle^{\otimes t} = (-1)^{|a \wedge b|}\, 2^{-|a|/2} \sum_{w \le a} i^{|w|} Z^{b+w} |T\rangle^{\otimes t}. \tag{S3}$$

Now apply Π_C. By Definition 10, the terms with $b + w \notin \Sigma \cup (\ell + \Sigma)$ vanish. If $e \in c\ell + \Sigma$, then $e + c\ell \in \Sigma$, $Z^{e+c\ell} = \xi(e + c\ell) g$ with $g \in S_C$, and $\Pi_C g = \Pi_C$. Since Z^ℓ commutes with S_C, it commutes with Π_C, so $\Pi_C Z^e |T\rangle^{\otimes t} = \chi(e + c\ell) Z^{c\ell} \Pi_C |T\rangle^{\otimes t}$. This proves Eq. (S1) with $\mu = (-1)^{|a \wedge b|}$. For the probabilities, note that the vector $Z^\ell \Pi_C |T\rangle^{\otimes t}$ has the same norm as $\Pi_C |T\rangle^{\otimes t}$. The isometry V maps the two to multiples, of equal modulus, of $Z|T\rangle$ and $|T\rangle$, which are orthogonal. The acceptance and the bad weight follow, and the bad weight is positive exactly when $g_1 \neq 0$. □

Proof of Proposition 12. We first write g_c as a sum over a quadratic form and then remove its variables one at a time. The set $\{w \le a : w + b \in c\ell + \Sigma\}$ is empty or an affine subspace $w_0 + \langle k_1, \dots, k_m \rangle$ of $\mathbb{F}_2^{|a|}$, which Gaussian elimination finds in $O(t^3)$. For binary vectors $v_0, \dots, v_m$, inclusion and exclusion give $|v_0 + \cdots + v_m| = \sum_{S \neq \emptyset} (-2)^{|S|-1} |\bigwedge_{s \in S} v_s|$, and the terms with $|S| \ge 3$ vanish modulo four. We write $\chi(w) = (-1)^{x \cdot w}$. Since $2(x \cdot w) \bmod 4$ is linear in z, the summand of Eq. (S2) is $i^{q(z)}$ with

$$q(z) = q_0 + \sum_l L_l z_l + 2 \sum_{l < l'} B_{ll'} z_l z_{l'}, \quad L_l = |k_l| - 2|w_0 \wedge k_l| + 2\, x \cdot k_l, \quad B_{ll'} = |k_l \wedge k_{l'}| \bmod 2. \tag{S4}$$

To sum out z_v, let N be the set of its neighbors, those with $B_{vl} = 1$, and let $y = \bigoplus_{l \in N} z_l$. The sum over z_v is $1 + i^{L_v}(-1)^y$. If L_v is odd, this is $(1 + i)\, i^{3y}$ or $(1 - i)\, i^y$. The parity y, read as an integer modulo four, is $\sum_{l \in N} z_l - 2 \sum_{l < l'} z_l z_{l'}$, so q stays quadratic. If L_v is even, the sum over z_v is $2[y = L_v/2]$, where the bracket is one when the condition holds and zero otherwise. If N is empty, the sum then vanishes or is multiplied by two. Otherwise the constraint expresses one variable of N as a parity of the others, which is substituted into q in the same way. Each step costs $O(m^2)$ and multiplies the modulus by $\sqrt{2}$ or 2, or returns zero. Hence $|g_c|$ is zero or a power of $\sqrt{2}$, and the acceptance $2^{-|a|}(|g_0|^2 + |g_1|^2)$ and the bad weight $2^{-|a|}|g_1|^2$ are dyadic. We checked the procedure against direct summation on 400 random forms with up to eight variables. □

S3. CULTIVATION AS DISTILLATION

This section proves Theorem 3. Its first half says that a circuit of fault distance d needs as many T gates as an order-d distillation protocol needs inputs, and it follows directly from Theorem 1. Its second half, that this number of T gates also suffices, needs a construction.

Corollary 13. *In a noise model in which every T gate may suffer a Z fault, every accepted run of a protocol of fault distance d executes at least $n_{\min}(\tau, d)$ T gates, the least number of noisy T inputs of any order-d distillation protocol for τ.*

Proof. We show that the code on the T gates defines a distillation protocol. By Proposition 7 it suffices to consider a correct non-adaptive circuit with t T gates and fault distance d. Configurations of Z faults at T gates have $D = 0$ and $E = Z^e$. By Theorem 1, each of them with $|e| < d$ therefore satisfies $\Pi_C Z^e |T\rangle^{\otimes t} \in \mathbb{C}\, \Pi_C |T\rangle^{\otimes t}$. Measuring the stabilizers of C on t noisy T states, postselecting and decoding with V is therefore an order-d distillation protocol [19]. □

The proof uses only the *order* d_C of the code. This is the least weight of a Z pattern that violates the distillation condition, and it is the suppression order of the distillation protocol that measures C on noisy T states. The Clifford part limits the fault distance in a second way, by spreading faults. We call $d_{\rm sp} = \min\{\mathrm{wt}(F) : D(F) = 0,\ |E(F)|_S > \mathrm{wt}(F)\}$ the *spreading distance* of the circuit. It is the least number of undetected faults whose Pauli operator on the T inputs is heavier than their number.

Theorem 14. *A correct non-adaptive circuit with code C has fault distance d with $\min(d_C, d_{\rm sp}) \le d \le d_C$. Conversely, for every stabilizer code C on t qubits that encodes a target τ with $\nu(\tau) = r$ as $\Pi_C|T\rangle^{\otimes t} \neq 0$, there is a correct non-adaptive circuit with t T gates and an admissible ideal decoder whose code on the T gates is C and whose fault distance is d_C.*

The upper bound is the proof of Corollary 13, and the lower bound is Lemma 8(ii). For the converse, proved below, the circuit applies T to t qubits in $|+\rangle$, so that its T inputs are its data. It then measures the generators of C one after another with cat states [23], each checked by $d_C - 1$ rounds of parity measurements. A cat state that passes a fault-free round has no X error, so k faults in one measurement leave a data error of weight at most k. The output stays encoded in C. The ideal decoder, which measures C again, rejects every run whose recorded signs disagree with the data error. Hence $d_{\rm sp} \ge d_C$.

Two limits of Theorem 14 should be kept in mind. First, the converse relies on the ideal decoder. It re-measures every generator, so it catches every wrong sign of the last noisy measurement and every data error left after it. Each of its projections repeats an earlier measurement with only commuting operations in between, so it is a detector of the skeleton and makes no check that depends on the T inputs, unlike the decoder of the published circuit (Sec. S9). This is allowed by the definition of an admissible decoder and is the convention of Ref. [1], but a real final stage must do this work with noisy operations. The output is also left in the code C itself, for example the 15-qubit Reed-Muller code, rather than in a code that is later grown into a surface code. The cost of that transfer is outside our model. The lower bound on the number of T gates, Corollary 13, has no such caveat, since an ideal decoder can only help a protocol. Second, the lower bound $\min(d_C, d_{\rm sp})$ does not make the fault distance cheap to compute. Finding $d_{\rm sp}$ still means enumerating undetected configurations up to weight d_C, and it only removes the collapse tests. For the published circuit of Proposition 5 it leaves $d \in \{2, 3\}$ (Sec. S4). The value of the theorem is structural. It splits the fault distance into the order of the code and the spreading of faults, and it shows that every code is attained.

The equality combines Corollary 13 with Theorem 14 for the code of an optimal distillation protocol, which has distance at least d by Lemma 8(ii). The last bounds follow from Rains' theorem that a quantum code of length n corrects at most $\lfloor (n+1)/6 \rfloor$ errors [35]. The exact values of Ref. [19] thus hold for cultivation: fault distance two needs nine T gates for one T state and eight for $|\mathrm{CCZ}\rangle$. The distance bound is a statement about distillation alone. By Lemma 8(ii), the code of every order-d distillation protocol for τ has distance at least d, so $n_{\min}(\tau, d)$ is at least the least length of such a code. We present it here as a new distillation result. It improves Table I of Ref. [19], for one T state from $d = 6$ on and for $|\mathrm{CCZ}\rangle$ from $d = 4$ on, and from $d = 7$ on it also improves the analytic bound $2d$ for one T state proved there. For codes with transversal T gates, distance and suppression order are known to be linked [36], and Kalra and Prakash bound suppression exponents as functions of the length for self-orthogonal codes over GF(4) [37]. We did not find the statement for arbitrary postselected protocols. The linear-programming bound for stabilizer codes with Rains' shadow enumerator [35, 38], certified in exact arithmetic (Table SI), raises these lower bounds. For one T state at $d = 6, \ldots, 12$ it raises them from $\max(2d, d+7)$ to 16, 17, 22, 23, 28, 29 and 34. For $|\mathrm{CCZ}\rangle$ at $d = 4, \ldots, 8$ it raises them from 11, 12, 13, 15 and 17 to 12, 14, 17, 20 and 23.

Lemma 15 (Checked cat-state measurement)**.** *Let a Pauli operator g act on w data qubits with letters $P_1, \ldots, P_w$. Prepare $(|0\rangle^{\otimes w} + |1\rangle^{\otimes w})/\sqrt{2}$ on cat qubits $c_1, \ldots, c_w$ by preparing c_1 in $|+\rangle$ and the others in $|0\rangle$ and applying CNOTs from c_1. In each of θ rounds, measure $Z_{c_1} Z_{c_i}$ for $i = 2, \ldots, w$ with a fresh ancilla in $|0\rangle$, the target of CNOTs from c_1 and c_i, and keep the outcome $+1$. Apply controlled-P_i from c_i to the ith data qubit for every i. Measure every c_i in the X basis and keep the runs whose outcomes have product s. If at most θ faults occur in this gadget and the run is kept, it acts on the data as $R\,(I + s'g)/2$ up to a nonzero scalar, with $s' = \pm s$ and a Pauli operator R for which R or Rg acts on at most as many qubits as there are faults.*

Proof. We track the X errors that the cat qubits carry onto the data. Without faults, X outcomes with product $(-1)^{|x|}$ leave a nonzero multiple of $[I + (-1)^{|x|} g]/2$ on the data. Now push every fault to the end of the gadget. Z errors on cat qubits, which Z errors on the ancillas also produce, and flips of the X measurements change the product of the X outcomes. X errors on the ancillas and flips of their measurements change only the checks. A Pauli on a data qubit before its controlled Pauli can be moved after it, at the cost of a Z error on its cat qubit when it anticommutes with P_i. Let $x \in \mathbb{F}_2^w$ mark the X errors on the cat qubits at the controlled Paulis. They reach the data as $P^x = \bigotimes_i P_i^{x_i}$, and $P^{x+\mathbf{1}} = \pm g P^x$. During the rounds the cat qubits are only controls of CNOTs, so a fault there changes at most

one entry of x. The same holds for a fault on a cat qubit before its own controlled Pauli. A fault in a controlled Pauli or on a data qubit adds a Pauli on one data qubit, since an X error on c_i after its controlled Pauli no longer spreads. If the preparation of the cat state is fault-free, x starts at $\mathbf{0}$. Otherwise at most $\theta-1$ faults fall into the θ rounds, so one round is fault-free. That round reads x_1+x_i correctly for every i, so a kept run has $x\in\{\mathbf{0},\mathbf{1}\}$ after it, and only later faults change x. In both cases the data error is R or $\pm Rg$, with R acting on at most as many qubits as there are faults. □

Proof of Theorem 14. We first prove the two bounds and then build the circuit of the converse.

Bounds. Choose a pattern e of weight d_C that violates the distillation condition. Z faults at the corresponding T gates have $D=0$ and $E=Z^e$ (Theorem 1). They are therefore harmful, and $d\le d_C$. If $\mathrm{wt}(F)<\min(d_C,d_{\mathrm{sp}})$ and $D(F)=0$, then $|E(F)|_S\le\mathrm{wt}(F)<d_C$, and F is harmless by Lemma 8(ii).

Construction. Let $g_1,\dots,g_m$ generate S_C, with signs s_k. The circuit prepares t data qubits in $|+\rangle$ and applies T to each. It then measures $g_1,\dots,g_m$ in turn with the gadget of Lemma 15, with $\theta=d_C-1$ and $s=s_k$. Finally it passes the data to the ideal decoder of C, which projects onto $(I+s_kg_k)/2$ for every k and unencodes. The circuit keeps every run whose cat-state outcomes have the required product. Fixing one such outcome string for each cat state gives the same Kraus operator up to a scalar, so the construction is a fully postselected circuit in the sense of Sec. S1. Every projection of the decoder onto g_k follows the measurement of g_k with only commuting operations in between, so it is fixed in the skeleton and is a detector. Without faults the checks of the cat states are deterministic and the measurements apply Π_C. The output is then a nonzero multiple of $\Pi_C|T\rangle^{\otimes t}$, so the circuit is correct and the decoder is admissible. In the skeleton, the gadget of a T gate on $|+\rangle$ maps its input to the data qubit identically, $\sqrt{2}\langle 0|_a\mathrm{CX}_{q\to a}|+\rangle_q|\psi\rangle_a=|\psi\rangle_q$, so the code of the circuit is C. Now let F have fewer than d_C faults, and let every check of the cat states pass. By Lemma 15, the kth measurement acts as $R_k(I+s'_kg_k)/2$, with R_k or R_kg_k acting on at most as many qubits as there are faults in that gadget. Every other fault, on a T gate or an idle data qubit, is a Pauli operator on one data qubit. So is a fault on the preparation of $|+\rangle$, which leaves $|+\rangle$ or $|-\rangle$, with $T|-\rangle=ZT|+\rangle$. We move all these Pauli operators to the T inputs. A Pauli R moved past $(I+s'g)/2$ turns it into $(I+(-1)^{\langle R,g\rangle}s'g)/2$, where $\langle R,g\rangle$ is zero if R and g commute and one otherwise. The circuit therefore applies $\prod_k(I+s_kg_k)/2\cdot\prod_k(I+s''_kg_k)/2\cdot R\,|T\rangle^{\otimes t}$ with signs s''_k, where R is the product of all the Paulis. Since the projections commute, this vanishes unless $s''_k=s_k$ for every k, and it is then a multiple of $\Pi_CR|T\rangle^{\otimes t}$. Replacing R_k by R_kg_k changes R only within $R\langle iI,S_C\rangle$, so $|R|_S\le\mathrm{wt}(F)<d_C$, and Lemma 8(ii) shows that F is harmless. Faults that fail a check are rejected. Hence the fault distance is at least d_C, and by the bound above it equals d_C. □

Proof of Theorem 3. We prove the equality first and the two lower bounds after it. By Corollary 13, every accepted run executes at least $n_{\min}(\tau,d)$ T gates. Conversely, by Theorem 1 of Ref. [19], an order-d distillation protocol on $n=n_{\min}(\tau,d)$ inputs yields a stabilizer code C on n qubits whose projection of $|T\rangle^{\otimes n}$ encodes τ and satisfies the distillation condition for all patterns of weight below d. So $d_C\ge d$, and Theorem 14 turns C into a circuit with n T gates and fault distance $d_C\ge d$. For the lower bounds, Lemma 8(ii) shows that C, and the code of every order-d distillation protocol, has r logical qubits and distance at least d. Such a code corrects $\lfloor(d-1)/2\rfloor$ errors. By Ref. [35] its length is therefore at least $6\lfloor(d-1)/2\rfloor-1$, which is $3d-4$ for odd d. For even d, feeding one input of an order-d protocol with $I/2$ gives an order-$(d-1)$ protocol [19], so $n_{\min}(\tau,d)\ge n_{\min}(\tau,d-1)+1\ge 3d-6$. □

S4. CODES ON THE T GATES

This section collects results about the code C on the T gates. We bound the length of the codes that a given fault distance needs, test the construction of Theorem 14 on three codes, and look at the code of a published circuit.

Linear-programming bound. By Theorem 3, lower bounds on the length of stabilizer codes are lower bounds on the number of T gates. Here we explain how we obtain the bounds of Table SI. Consider an $[[n,r]]$ stabilizer code with stabilizer group S. Let A_j be the number of elements of weight j of S, and B_j the number of elements of weight j of its normalizer, both counted up to phases. Let S_j be the number of Pauli operators of weight j in the shadow of the code, the set of Pauli operators that commute with the elements of even weight of S and anticommute with those of odd weight. These numbers satisfy [35, 38]

$$A_0=1,\qquad \textstyle\sum_j A_j=2^{n-r},\qquad B_j=2^{r-n}\sum_i K_j(i)A_i,\qquad S_j=2^{r-n}\sum_i(-1)^iK_j(i)A_i,$$
$$B_j\ge A_j\ge 0,\qquad S_j\ge 0\qquad (0\le j\le n),$$

where $K_j(i)=\sum_s(-1)^s3^{j-s}\binom{i}{s}\binom{n-i}{j-s}$ is the coefficient of y^j in $(1+3y)^{n-i}(1-y)^i$. A code of distance d also satisfies $B_j=A_j$ for $1\le j<d$. If no real A satisfies this system, there is no $[[n,r,d]]$ stabilizer code. There is then none of

smaller length either, since appending a qubit in $|0\rangle$, with stabilizer Z, keeps the distance. For each entry of Table SI, the script `scripts/cultivation_codes.py` (option `--lp`) shows that the system has no solution at length $n-1$. It computes a Farkas certificate of infeasibility, a combination of the constraints that leads to a contradiction, by an exact phase-one simplex method in rational arithmetic, and checks the certificate in integer arithmetic. As a test, the weight distributions of the nine-input, eight-input and 15-input codes satisfy the system. For $r = 1$ and odd d the resulting bounds coincide with Rains' bound $3d - 4$.

TABLE SI. Lower bounds on the length n of a stabilizer code with r logical qubits and distance d, from the linear-programming bound with shadow, certified in exact arithmetic, and the corresponding lower bounds on $n_{\min}$ of Ref. [19]. By Theorem 3, $n_{\min}$ is at least the larger of the two, both for one T state ($r = 1$) and for $|\mathrm{CCZ}\rangle$ ($r = 3$).

$r = 1$, d	6	7	8	9	10	11	12
$n \geq$	16	17	22	23	28	29	34
Ref. [19]	13	14	16	18	20	22	24
$r = 3$, d	4	5	6	7	8		
$n \geq$	12	14	17	20	23		
Ref. [19]	11	12	13	15	17		

The construction. We tested the construction of Theorem 14 on three codes. The same script builds the circuit for each code, treats every operation as a noisy location, and computes the fault map. Pairs of faults are enumerated through their detector vectors. Since D is linear, a pair is undetected exactly when its two faults have the same D. The first two codes are the eight-input CCZ code, with 8 T gates, 17 qubits, 97 two-qubit gates and one round of checks, and the nine-input T code, with 9 T gates, 16 qubits, 128 two-qubit gates and one round. None of the 1860 single faults of the first circuit and none of the 2450 single faults of the second is harmful. Both circuits therefore have fault distance two, the order of their codes. The third code is the 15-to-1 code. With two rounds of checks (15 T gates, 24 qubits, 458 two-qubit gates), none of the 8612 single faults and none of the 37,028,662 pairs is harmful. With one round (310 two-qubit gates), none of the 5874 single faults and none of the 17,214,903 pairs is harmful. Without faults, the three circuits accept with the probabilities 1/16, 1/128 and 1/1024 with which their codes accept ideal T states, and every projection of their decoders is a detector. The script also checks Lemma 8(ii) for these codes: all Pauli operators that act on fewer than d_C inputs collapse, 27 of them for the nine-input code, 24 for the eight-input code and 990 for the 15-to-1 code.

The published circuit. The code of a published circuit shows that a heavy Pauli operator on the T inputs is needed for harm but does not cause it by itself. We take the circuit of Proposition 5 with 15 T gates, the distance-three circuit of Ref. [1]. All Z patterns of weights one and two on its T gates collapse, and 35 of weight three do not, so $d_C = 3$. Undetected single faults have $|E|_S \leq 1$. Among the undetected pairs of faults at distinct locations, 10,942, 12,338, 3887, 60 and 301 have $|E|_S = 0, \ldots, 4$, and all of these are harmless. Another 87 have $|E|_S = 5$, and 48 of them are harmful. A Pauli of weight at least d_C modulo S_C is thus necessary for harm but not sufficient.

S5. THE SMALLEST CASE

By Theorem 3, fault distance two for one T state needs exactly nine T gates. This section gives a small circuit that attains this bound, proves its fault distance, and describes the integer program that found it and its logical error under depolarizing noise. The construction of Theorem 14 with the nine-input factory code of Ref. [19] also uses nine T gates, but it needs 16 qubits and 128 two-qubit gates. An integer program over the fault map finds a smaller circuit.

Proposition 16. *There is a circuit with nine T gates, eleven qubits and* 34 *two-qubit gates, ending in an admissible ideal decoder, that prepares one T state with fault distance two.*

The circuit (Fig. S1(b)) prepares nine qubits in $|+\rangle$ and applies T to each, so that its T inputs are its data qubits. It then measures eight independent stabilizers of weight four of the nine-input factory code of Ref. [19], each with one ancilla and four controlled-Pauli gates. It keeps only the outcomes of the factory, which occur with probability 1/128 without faults. The factory code has distance three. The output stays encoded in it, and the decoder, which measures the stabilizers again, is admissible.

The argument for fault distance two sorts the faults by where they occur. Faults on the preparations and T gates are single-qubit Paulis on the inputs. They collapse because the code distills to order two, by Lemma 8(ii). A fault

during the measurement of the kth stabilizer leaves a Pauli R on the data and may flip the outcome. It is detected if R anticommutes with a stabilizer measured before, or if the flip does not match the commutation of R with the kth stabilizer. Otherwise it acts as R on the inputs, and every Pauli of weight two on the inputs modulo the stabilizers is harmful (see the integer program below). The weight-two errors that the measurement circuits spread to the data must therefore anticommute with a stabilizer measured earlier, or be caught by a flag [39], an extra qubit that detects errors of the ancilla that would spread to two data qubits. The choice of the eight stabilizers, their order, the order of the qubits in each measurement and the flags is an integer program over the fault map, described below. The first measurement needs a flag, since no stabilizer precedes it. Gurobi shows in under a minute that one flag suffices. The proof is exact: the remaining single faults reduce to Paulis on the inputs of weight at most one modulo the stabilizers. An enumeration of all 627 single faults confirms it, both with the fault map and by direct simulation (Fig. S1(c)).

The construction is a distillation at the physical level, in the spirit of Ref. [10], made fault tolerant by the order of its measurements. Idle faults do not lower its fault distance, since an idle fault equals a fault after the preceding operation on the same qubit. The circuit minimizes T gates, not cost. It accepts one fault-free run in 128. Under the uniform depolarizing noise of Ref. [1], its logical error per accepted run is $53p^2$ to leading order without idle noise, and $1.9 \times 10^3 p^2$ with idle noise in an as-soon-as-possible schedule (see below). For comparison, the distance-three circuit of Ref. [1] has the same fault distance with 15 T gates, 15 qubits and 81 CNOTs. It accepts every fault-free run and has error $32p^2/75$.

For fault distance three, Theorem 14 applied to the 15-to-1 code of Bravyi and Kitaev [40] gives a circuit with 15 T gates, 24 qubits and 310 two-qubit gates (Sec. S4). With the lower bound of Ref. [19], the least number of T gates for fault distance three is therefore between ten and fifteen. Like the nine-T circuit, this circuit is built to use few T gates, not to be cheap. Without faults it accepts one run in 1024, because the outcomes of its stabilizer measurements are random and only one outcome string is kept. The distance-five circuit of Ref. [1] reaches the same fault distance with real T gates using 53 T gates, 42 qubits and 477 two-qubit gates, and it accepts every fault-free run.

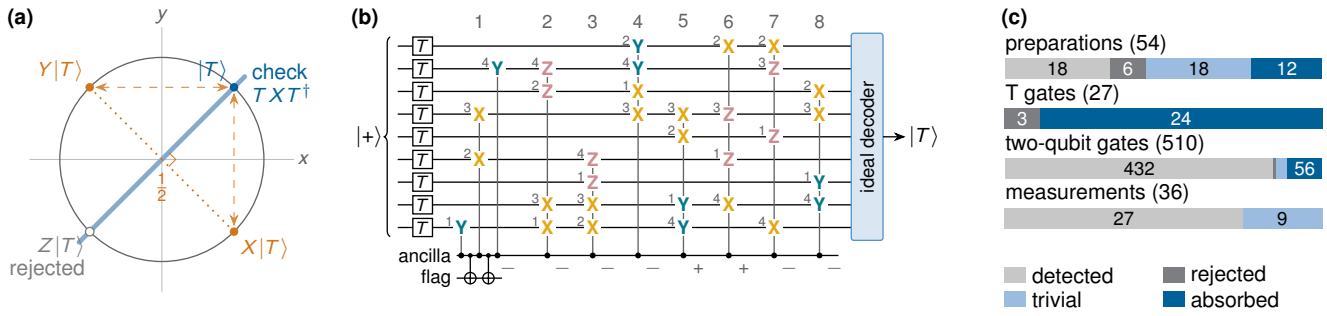


FIG. S1. Fault distance two with nine T gates. (a) A check that projects onto $|T\rangle$, drawn on the equator of the Bloch sphere. A Z fault maps $|T\rangle$ to the antipodal, orthogonal state $Z|T\rangle$, which the check rejects. X and Y faults reflect $|T\rangle$ in the x and y axes. The reflected states lie at right angles to the check axis, so the check accepts them with probability $1/2$ and then outputs exactly $|T\rangle$. Faults of this kind are harmless although no detector sees them. (b) The circuit of Proposition 16. Nine qubits are prepared in $|+\rangle$ and receive one T gate each. Eight stabilizers of the nine-input factory code are then measured in turn (columns 1 to 8). Each column couples the ancilla by controlled-Pauli gates to the four marked qubits, with the marked Pauli operators, in the order given by the small numbers, and keeps the indicated outcome. The first measurement carries a flag, coupled to the ancilla after the first and before the last of these gates, which detects errors of the ancilla that would spread to two data qubits. The output remains encoded in the factory code, which has distance three, and an ideal decoder measures its stabilizers and unencodes. (c) The 627 single faults of the circuit, by location type. Each fault is detected (a flag, or a stabilizer measured both before and after it, changes its outcome), rejected (its Pauli on the T inputs projects them to zero), trivial (that Pauli is a stabilizer of the code on the T gates) or absorbed (the projection is a nonzero multiple of the encoded T state, as in panel a). The narrow segments of the two-qubit gates are 4 rejected and 18 trivial faults. No single fault is harmful.

Proof of Proposition 16. Let $g_1, \ldots, g_8$ be the measured stabilizers in their order (Table SII). They generate the stabilizer group S of the factory code, which is the code C of Theorem 1 for this circuit, with the T gates numbered by their qubits. A fault after a preparation or a T gate is, up to stabilizers of $|+\rangle$, a Pauli of weight one on the T inputs, with no flips. Its Z part collapses, by the exact certificate of the single-fault minors in Ref. [19], and so does its X part, by Lemma 8(i). Consider a fault in the measurement of g_k, including its ancilla and flag. Pushed to the end of the measurement circuit, it becomes a Pauli R on the data, a flip ϕ of the outcome of g_k and a flip ψ of the flag. If $\psi = 1$, the flag detects it. If R anticommutes with some g_j, $j < k$, the decoder detects it, because g_j was projected before R and is projected again after it, with only commuting projections in between. If ϕ differs

from the commutator $\langle R, g_k\rangle$, the decoder detects it, because the projection onto g_k then has the wrong sign relative to the decoder's. Otherwise R commutes with $g_1, \ldots, g_{k-1}$ and, together with the flip, equals R applied before the measurement of g_k. It can then be moved to the inputs, and the fault is harmless exactly when R collapses. An exact computation over $\mathbb{F}_2$ (`scripts/cultivation_t9.py`) sorts the 573 faults in the measurement circuits of Table SII: 37 raise the flag, 288 leave an R that anticommutes with an earlier stabilizer, 152 flip the outcome inconsistently with R, and the remaining 96 leave an R of weight at most one modulo S, which collapses, since multiplication by a stabilizer changes $\Pi_C R|T\rangle^{\otimes 9}$ only by a sign. Measurement flips and faults on the decoder's input are covered by the same cases. □

Table SII lists the circuit of Proposition 16 in full. Qubits 0 to 8 are prepared in $|+\rangle$ and receive a T gate each. The kth measurement prepares the ancilla in $|+\rangle$, applies controlled-P gates from the ancilla to the listed qubits in the listed order, with P the letter of g_k on that qubit, and measures the ancilla in the X basis. It keeps the outcome equal to the sign of g_k. In the flagged measurement, a flag qubit in $|0\rangle$ is the target of CNOTs from the ancilla after the first and before the last controlled-Pauli gate. It is measured in the Z basis, and the outcome +1 is kept. The ideal decoder measures $g_1, \ldots, g_8$ with their signs and unencodes. The circuit has 18 preparations, 9 T gates, 34 two-qubit gates and 9 noisy measurements, 70 locations in all.

TABLE SII. The measurements of the nine-T circuit, in order: the stabilizer of the nine-input factory code (qubits 0 to 8 from left to right), the order of the controlled-Pauli gates, and the flag.

k	g_k	order	flag
1	$-IYIXIXIIY$	8, 5, 3, 1	yes
2	$-IZZIIIIXX$	8, 2, 7, 1	
3	$-IIIIIZZXX$	6, 8, 7, 5	
4	$-YYXXIIIII$	2, 0, 3, 1	
5	$+IIIXXIIYY$	7, 4, 3, 8	
6	$+XIIZIZIXI$	5, 0, 3, 7	
7	$-XZIIZIIIX$	4, 0, 1, 8	
8	$-IIXXIIYYI$	6, 2, 3, 7	

Integer program. The circuit was found by an integer program that chooses the stabilizers, their order, the order of the gates in each measurement and the flags. Let $\mathcal{W}$ be the 18 elements of weight four of the stabilizer group S of the factory code, with their signs, which span S. Let $\langle\cdot,\cdot\rangle$ be the symplectic form, and let $\mathcal{C}$ be the 48 minimal linearly dependent subsets of $\mathcal{W}$. For $w \in \mathcal{W}$, an order π of its support and $f \in \{0,1\}$, the measurement circuit (w, π, f) is that of Table SII, with a flag if $f = 1$. Let $\mathcal{R}(w, \pi, f)$ be the set of data Paulis R of the single faults in this circuit that raise no flag, satisfy $\phi = \langle R, w\rangle$ and do not collapse. By the proof of Proposition 16, these are the faults that are harmless only if R anticommutes with a stabilizer measured earlier. They are computed once, by Pauli propagation and the collapse test on nine T states. Every Pauli that collapses has weight at most one modulo S. Binary variables s_w select the measured stabilizers, y_{vw} equals one if v is measured before w, and $z_{w\pi f}$ selects the order and the flag of the measurement of w:

$$
\begin{aligned}
\min_{s,y,z}\quad & \sum_{w\in\mathcal{W}}\sum_{\pi} z_{w\pi 1} \\
\text{s.t.}\quad & \textstyle\sum_{w\in\mathcal{W}} s_w = 8, \qquad \sum_{w\in C} s_w \le |C| - 1 \quad (C \in \mathcal{C}), \\
& \textstyle\sum_{\pi,f} z_{w\pi f} = s_w \quad (w \in \mathcal{W}), \\
& y_{vw} \le s_v, \quad y_{vw} \le s_w, \quad y_{vw} + y_{wv} \le 1, \quad y_{vw} + y_{wv} \ge s_v + s_w - 1 \quad (v \ne w), \\
& y_{uv} + y_{vw} + y_{wu} \le 2 \quad (u, v, w \text{ distinct}), \\
& z_{w\pi f} \le \textstyle\sum_{v\in\mathcal{W}:\,\langle R,v\rangle=1} y_{vw} \quad (R \in \mathcal{R}(w,\pi,f)), \\
& s_w,\ y_{vw},\ z_{w\pi f} \in \{0,1\}.
\end{aligned}
$$

The objective counts the flags. The first line of constraints selects eight independent stabilizers, which then generate S, since a subset of $\mathcal{W}$ is independent exactly when it contains no set of $\mathcal{C}$. The second line picks one order and one flag setting for each selected stabilizer. The constraints on y make it a linear order of the selected stabilizers, since a tournament without directed triangles is transitive. The constraint on z and $\mathcal{R}$ requires every dangerous Pauli, that is,

every element of $\mathcal{R}(w, \pi, f)$, to anticommute with a stabilizer measured before. The program has 1188 variables and 4777 constraints. Gurobi 10 finds a design with one flag in 46 s. This design is optimal, because the first measurement needs a flag. The script `scripts/cultivation_t9.py` builds the program (option `--mip`), verifies the circuit, and writes `results/cultivation/t9_circuit.json`. Of the 627 single faults, 477 are detected, 13 rejected, 45 trivial and 92 absorbed (Fig. S1(c)). Of the 192,546 pairs, 7146 are harmful.

Depolarizing noise. We also computed the logical error of the circuit under depolarizing noise. Under the uniform depolarizing model of Sec. S9 without idle noise, 6279 pairs of faults with nonzero probability are harmful. The logical error per accepted run is $c_2 p^2 + O(p^3)$ with $c_2 = 53.1$. Pairs of faults on the preparations and T gates, which act as faults of the factory's inputs, contribute 576 of the harmful pairs. To include idle noise, we schedule every operation as early as possible, except that the ancilla and flag are prepared just before their first gate. The circuit then has 411 idle locations, 409 of them on the data. With idle noise, 132,968 pairs are harmful and $c_2 = 1.9 \times 10^3$. Most of this comes from pairs of idle faults on the data, which wait while the eight stabilizers are measured one after another.

S6. THE SINGLE-QUBIT CHECK

This section works out the smallest example of Theorem 1 by hand. The smallest circuit with a check prepares $|+\rangle$, applies the injection T gate, then $\mathrm{T}^\dagger$, projects onto $X = +1$ and applies T. Its fault-free output is $|T\rangle$, since $|T\rangle$ is the $+1$ eigenstate of $TXT^\dagger$. In the skeleton the three T gates become ideal inputs a_1, a_2, a_3. The first two gadgets project the data qubit onto the same computational-basis state $|b\rangle$, and the $S^\dagger$ of $\mathrm{T}^\dagger = S^\dagger T$ contributes the phase $(-i)^b$. The skeleton therefore projects $a_1 a_2$ onto $|00\rangle + i|11\rangle$ and passes a_3 to the output. The code C has stabilizers Z_1Z_2 and X_1Y_2, and $\langle 00| + (-i)\langle 11|$ applied to $|T\rangle^{\otimes 2}$ gives $(1 - i\omega^2)/2 = 1$. Table SIII lists every single fault. No fault triggers a detector of the skeleton. The check rejects Z-type errors, for which the projection in the collapse condition is zero. It projects X-type errors after the injection back onto $|T\rangle$, as in Fig. S1(a). Faults after the last T gate are harmful, since nothing protects the output.

TABLE SIII. Single faults of the single-qubit check. E is the Pauli operator on the inputs $a_1 a_2 a_3$, given as a representative modulo $\langle iI, S_C\rangle$ with $S_C = \langle Z_1Z_2, X_1Y_2\rangle$. The last column is the acceptance probability relative to the fault-free run.

location	fault	E	harmless	acceptance
after preparation of $\|+\rangle$	X	III	yes	1
	Y, Z	IZI	yes (rejected)	0
after the injection T	X	IYI	yes	1/2
	Y	IXI	yes	1/2
	Z	IZI	yes (rejected)	0
after $\mathrm{T}^\dagger$, before the projection	X	III	yes	1
	Y, Z	IZI	yes (rejected)	0
projection	sign flip	IZZ	yes (rejected)	0
after the last T	X, Y, Z	IIX, IIY, IIZ	no	1

S7. COMPUTING THE FAULT MAP

We implement Theorems 1 and 2 twice, independently. This section describes the first implementation, the script `scripts/cultiv.py`. It computes the maps D and E symbolically and decides the collapse condition with state vectors, so it serves for small circuits. The second implementation, built on stim and Gauss sums, is the subject of Sec. S8.

The script `scripts/cultiv.py` takes a non-adaptive circuit given as a list of operations: preparations, Clifford gates, T and $\mathrm{T}^\dagger$ gates, projections (with or without reset of the measured qubit), discards of qubits in known eigenstates, and identity operations that carry idle noise. As in the proof of Theorem 1, it follows the Choi state of the skeleton on the physical qubits, the gadget inputs a_j and their references r_j. The stabilizer generators are stored as binary vectors. The sign of each generator is stored as a linear form in the flips σ of all projections. This is all that the maps need, since constant signs only affect phases. A projection is handled in one of two ways. If its Pauli

anticommutes with a generator, it replaces that generator, with the form σ_k of the kth projection. If its Pauli is in the group, it adds the detector row $\sigma_k + \sum_i \ell_i$, where ℓ_i are the forms of the generators that multiply to it.

For a fault F, Pauli propagation through the skeleton gives the flips $s(F)$ and the output Pauli $P_{\text{out}}(F)$. Then $D(F)$ is the value of the detector rows on $s(F)$. The generators whose forms evaluate to one on $s(F)$ change sign. The destabilizers, the Pauli operators that each anticommute with exactly one generator, convert these sign changes into a Pauli G on the physical qubits and references. Multiplying GP_{out} by the stabilizer with the same physical part, found by elimination, leaves a Pauli on the references. This Pauli is $E(F)$. It is reduced modulo S_C by a fixed echelon basis, so that the reduction is linear. When the circuit does not map onto its whole output space, as when the output is encoded and decoded, part $U(F)$ of the physical Pauli may remain unmatched. It is kept, and it vanishes whenever $D(F) = 0$ for the circuits of Theorem 1. The triple (D, U, E) of a configuration is the sum of those of its faults.

For $D(F) = 0$, harmlessness depends only on (U, E), and one simulation decides it. A Pauli $E = X^u Z^v$ on the T inputs has the same effect as replacing T^k by T^{k-2} at the gates in u, which is what the gadget does with input $X|T\rangle$, and applying Z after the gates in v. The circuit so modified, followed by U, is simulated with state vectors, and its output is compared with the fault-free one. Pairs of faults are enumerated through the classes of their triples, so that each class is simulated once.

The script `scripts/cultiv_test.py` tests this method against direct simulation of the faulty circuit. It uses 300 random circuits of two or three qubits with up to three T gates and two projections, with all single faults and random pairs, 42,367 configurations in all. Harmlessness from the fault map agrees with direct simulation on every one. For the 65 circuits whose skeleton map has full row rank, the script also computes the skeleton map M_F densely, by feeding basis states to the gadgets. It checks for 9,271 configurations that $M_F = 0$ exactly when $D(F) \neq 0$, and that otherwise $U(F) = 0$ and $M_F \propto ME(F)$.

A further script, `scripts/cultivation_check.py`, tests Theorem 1 with its own state-vector code. For all single faults and up to 800 pairs of faults, it compares the faulty output with physical T gates with the output of the skeleton fed with ideal T states, which tests the gadget identity. It then checks that the skeleton map is zero or a multiple of ME for a Pauli E on the T inputs, that harmlessness computed directly agrees with the collapse condition, that $E = Z_j$ for Z faults at T gates, and that $E(F_1 + F_2) = E(F_1)E(F_2)$ modulo $\langle iI, S_C\rangle$. We ran it on circuits for $\mathrm{CS}|{+}{+}\rangle$ and $|\mathrm{CCZ}\rangle$ and on ten random circuits on two qubits with two to four T gates and one or two projections. The gadget identity holds for all 10,542 configurations tested. In the six circuits whose skeleton map has full row rank, as Theorem 1 assumes, the other statements hold for all 5,316 configurations, all 24 T gates and all 3,726 pairs tested.

S8. EXACT EVALUATION AT SCALE

This section describes the second implementation, which handles large circuits. The circuits of Propositions 5 and 6 have up to 254 qubits and 77 T gates, beyond the reach of state vectors. We evaluate them with an independent implementation built on stim [20]. In it, the fault map of Theorem 1 is a detector error model, and the collapse condition is decided by the Gauss sums of Theorem 2.

In outline, one stim detector error model of the skeleton lists every fault class with its probability, its detector pattern D and its error E. One Walsh-Hadamard transform finds Σ and ℓ and checks Definition 10, after which every evaluation is a Gauss sum. The fault distance follows from a meet-in-the-middle search on detector patterns, and the logical error from sampling the error model with the exact weight of each sample.

Skeleton and Choi state. Every T gate on qubit q becomes CX_{q,a_j} followed by a measurement of a_j in the Z basis. A $\mathrm{T}^\dagger$ gate becomes the same followed by $S_q^\dagger$. Here a_j is half of a noiseless Bell pair with a reference r_j. Noise channels stay where they were, and the gadgets, Bell pairs and final decoding are noiseless. A noiseless tableau run fixes the reference outcomes. Every outcome that is random in the skeleton is fixed so that the circuit's own detectors read zero, and the remaining free outcomes are chosen. The state left on the references and the output qubit is the Choi state $|\Phi\rangle$ of $V\Pi_C$, with $t + 1$ stabilizer generators g_i.

Detectors and observables. The detectors are a basis of the parities of measurement records that are deterministic in the noiseless skeleton. We find them as the GF(2) null space of noiseless samples. The observables are the g_i, measured at the end, each completed by the record parity that makes it deterministic. One detector error model of the skeleton then gives every fault class with its probability p_c, detector pattern D_c and observable flips o_c. For $D = 0$, the effect of the faults on the Choi state is a Pauli R on the references whose commutation pattern with the g_i is o. A fixed right inverse over GF(2) gives R from o, and $R = E(F)$ up to transposition.

Exact evaluation. Row reduction of the g_i writes the Choi state as a sum over an affine subspace, $|\Phi\rangle = \sum_{x \in x_0 + \mathrm{span}(u_1,\ldots,u_\kappa)} c(x)|x\rangle$ with $c(x_0) = 1$ and $c(x + u_i) = i^{\varphi_i}(-1)^{z_i \cdot x} c(x)$. Here x_{out} and x_{ref} below denote the

output and reference bits of x. For $R = X^a Z^b$ the output is

$$v_y = \sum_{x:\, x_{\rm out}=y} c(x)(-1)^{b\cdot x}\omega^{|x_{\rm ref}\oplus a|}, \tag{S5}$$

and the acceptance and bad weight of the fault class are $|v|^2/|v_0|^2$ and $|v|^2/|v_0|^2 - |\langle v_0, v\rangle|^2/|v_0|^4$. A Z pattern e acts through $\beta(e) = (e\cdot u_1, \ldots, e\cdot u_\kappa) \in \mathbb{F}_2^\kappa$ and the sign $(-1)^{e\cdot x_0}$. One Walsh-Hadamard transform of length 2^κ over the coordinates therefore gives the output of every Z pattern. Definition 10 holds when exactly two values of β give a nonzero output, 0 and $\beta(\ell)$. Then $\Sigma = \beta^{-1}(0)$ and $\chi(e) = (-1)^{e\cdot x_0}$, and the conditions $w + b \in c\ell + \Sigma$ of Eq. (S2) become the linear equations $\beta(w) = \beta(b) + c\beta(\ell)$. Every further class is evaluated with the Gauss sums of Proposition 12. The direct sum above, of cost 2^κ, serves as a check. The affine dimension is $\kappa = 5$ for the $\delta = 3$ circuit of Ref. [1], 15 for its $\delta = 5$ circuit, 11 for the one-round cat checks at $\delta = 5$, 21 for two rounds and 20 at $\delta = 7$.

Searches and rates. The fault distance is the least k for which some k fault classes have total detector pattern zero and a harmful total o. We call k the weight of the search. Detector patterns are compressed by a random GF(2)-linear 64-bit hash. Weight three uses a sorted table of single hashes and checks every hit on the exact patterns. Weight four groups pairs of classes by hash and evaluates every distinct observable mask, the total o, of two pairs in the same group. A hash collision can only add masks. The certificates that find no harmful mask are therefore exact, and every harmful mask found is confirmed by a 4-set whose exact detector patterns cancel. Weight five splits each configuration canonically into its three smallest and two largest classes. The logical error at moderate p comes from sampling the detector error model, with the exact acceptance and bad weight of each sampled class. For rare failures we use the decomposition $P_{\rm bad} = \prod_c (1-p_c) \sum_k A_k$ with $A_k = \sum_{|S|=k} \prod_{c\in S} r_c\, [D_S = 0]\, {\rm bad}(o_S)$ and $r_c = p_c/(1-p_c)$. We evaluate A_k by drawing $k-2$ classes with probability proportional to $\prod r_c$ and summing exactly over all completing pairs. On the $\delta = 3$ circuit this agrees with the sampled logical error within statistical error.

Validation. We validated the second implementation on the $\delta = 3$ circuit of Ref. [1] with real T gates. A state-vector simulation inserted each of the 2,444 single physical faults and 400 random pairs explicitly and postselected on the reference outcomes. It agrees with the algorithm in rejection for all 2,224 rejected configurations, and in acceptance and bad weight to 3×10^{-16} for the 620 accepted ones. All 96 malignant pairs agree to the same precision. Each has acceptance 1/4 and bad weight 1/8 or 1/4, according to whether its output has infidelity 1/2 or 1. These are the 96 harmful pairs that the first implementation finds under the same uniform noise with idle errors (Sec. S9). They reduce to its 48 pairs when only the operations of the circuit carry faults, with the same leading coefficient 32/75. With S in place of T, the sampled acceptance and logical error at $p = 3\times 10^{-3}$ agree with direct stim sampling of the proxy circuit. The acceptance is 0.3251 in both, and the logical error is $1.08(13)\times 10^{-5}$ against $1.02(12)\times 10^{-5}$. The Gauss sums agree with the direct sum on random sums of one to four fault classes, all to 5×10^{-16}. We tested 335 classes of the $\delta = 3$ double check, 380 of the $\delta = 5$ protocol, 356 of the one-round verified cat at $\delta = 5$ and 60 of the unverified cat at $\delta = 7$. At $\kappa = 20$ the direct sum takes 0.22 s per class and the Gauss sums 1 ms. Every circuit that we study has deterministic checks. On one workstation, certifying fault distance five for the one-round checks at distance five takes seconds, and the weight-four certificates at distance seven take minutes.

S9. THE PUBLISHED CIRCUITS

This section gives the details behind part (i) of Proposition 5, for the distance-three and distance-five circuits of Ref. [1]. With T gates, the distance-three circuit has fault distance two [15, 16]. Theorem 1 recovers this and decides every class of faults exactly.

The distance-three circuit consists of a unitary injection followed by one double check. We compose it from the published chunks [22] (15 qubits, 15 T gates, 81 CNOTs) and place T gates where its stabilizer simulations use S gates. As in Ref. [1], we keep the final stabilizer measurements as an admissible ideal decoder, and we treat every operation as a noisy location. Then no single fault is harmful, and 48 of the 966,391 pairs are, while with S gates no pair is harmful. Fig. 2(a) shows one of the 48 pairs. Its mechanism was identified in Ref. [16]. The last layer of T gates turns an X error on two qubits into $[(X+Y)/\sqrt{2}]^{\otimes 2}$. The final X-type checks act only through C. They remove the components with Z parts, but not $X\otimes X$. Direct simulation confirms every harmful pair. In the language of Theorem 14, the loss comes from the spreading of faults, not from the code: the code on the 15 T gates has order $d_C = 3$, so a circuit with the same code reaches fault distance three, but the published circuit has spreading distance two (Sec. S4).

We use the uniform depolarizing noise of Ref. [1], with idle noise in every time step of the chunks. Under this noise 96 pairs are harmful. Write ϵ_T and ϵ_S for the logical error per accepted run with T and with S gates. Then

$\epsilon_T = 32p^2/75 + O(p^3)$, while $\epsilon_S = O(p^3)$ (derived below). The coefficient equals the exact one of Wan and Zapirain [15], who contracted a tensor network for an independently compiled circuit. The state-vector simulations published with the circuit [22], extrapolated linearly to $p = 0$, give $(\epsilon_T - \epsilon_S)/p^2 = 0.45 \pm 0.08$ (Fig. 2(b)). At $p = 10^{-3}$ the measured difference, $(5.8 \pm 0.6) \times 10^{-7}$, is more than half of ϵ_T, and at smaller p it dominates.

Reconstruction. We rebuilt the distance-three circuit from the published code. Its repository defines the circuit of Proposition 5 as two stim chunks, the unitary injection with one round of stabilizer measurements and the distance-three double-cat check, whose boundary conditions are measurements of the color-code stabilizers and of the logical Y. In the chunks, S and S_DAG stand for T and $\mathrm{T}^\dagger$. We joined the two chunks (`data/gsj_cultivation/`) by the coordinates of their qubits, dropped the boundary measurements between them, kept the final stabilizer measurements as an ideal decoder, and replaced S and S_DAG by T and $\mathrm{T}^\dagger$. Every measurement of the resulting circuit is deterministic, with T and with S, and the fault-free circuit accepts with probability one, so the decoder is admissible. With T gates, 4 of the 20 projections are not deterministic in the skeleton: the X measurement of the check and the three X-type stabilizer measurements of the decoder. They act through the code C, so the ideal decoder takes part in rejecting faults, as does any decoder that postselects on all stabilizers. Wan and Zapirain postselect in the same way, with a noiseless final projection onto every face stabilizer [15], and the agreement of the coefficient below suggests that the published state-vector simulations [22] do too. Exchanging T and $\mathrm{T}^\dagger$ throughout gives the same 48 harmful pairs. Grouped into fault classes, that is, single faults with the same detector pattern and the same Pauli operator on the T inputs, they form 9 harmful pairs of classes, with and without idle noise (`scripts/cultivation_gsj_classes.py`). These are the nine that the exact evaluation of Sec. S8 finds. Chan *et al.* group the error events of their own compilation by the signs they flip and the Pauli operator they leave at the end, and count the configurations whose output has fidelity zero. They find four of two faults [16]. Grouped their way, our 48 pairs form 12 classes, four with fidelity zero, as in Ref. [16], and eight with fidelity one half (`scripts/cultivation_codes.py`). The fault distance agrees with Ref. [16], and the leading coefficient with the independent exact result of Wan and Zapirain. The repository composes the chunks with its own compiler and re-layers the result. We did not run that code, so our idle noise follows the time steps of the chunk text.

Noise. We now derive the coefficient 32/75. With one location per operation, every Pauli error at every preparation, gate and measurement, and every flip of a measurement, is a fault. This gives 1397 faults in all on 137 operations. We use the uniform depolarizing model of Ref. [22] with strength p. It applies single-qubit depolarization after T (S) gates and two-qubit depolarization after CNOTs. It applies a bit flip after a Z reset and a phase flip after an X reset, and it flips every measurement result. Finally it applies single-qubit depolarization to every qubit that is idle in a time step. We apply this idle noise only to qubits between preparation and measurement, since noise on other qubits is removed by the next reset. This gives 436 locations, 299 of them idle. The logical error per accepted run is $\epsilon = \sum_F w_F h_F / \sum_F w_F \operatorname{Tr} \Lambda(\rho_F)$. Here w_F is the probability of the fault configuration F, and h_F is the probability that run F is accepted with a wrong output. The denominator is $1 + O(p)$. The coefficient of p^2 in the numerator is a sum over the 96 harmful pairs. These are 24 pairs of two CNOT faults, 24 of a CNOT and a T fault, 32 of an idle and a CNOT fault, 8 of an idle and a T fault, and 8 of two idle faults. Each pair contributes the product of its probabilities divided by p^2, times its relative acceptance (1/4 for each) and the infidelity (1/2 or 1) of its output. The sum equals 32/75. This is the leading coefficient of the exact series of Wan and Zapirain [15], whose circuit was compiled independently and has 518 fault locations. The coefficient therefore does not depend on our layering of the idle noise. The S version has no harmful pair.

Comparison. We compare this coefficient with the published simulations. Table SIV lists the state-vector simulations of Ref. [22] for this circuit, with T and with S gates at the same positions. As a consistency check, we fit $(\epsilon_T - \epsilon_S)/p^2 = a + bp$ by weighted least squares. The linear term is the next order of both series, and the fit only tests whether the data are compatible with the exact limit. The fit gives $a = 0.44 \pm 0.10$ for $p \le 3 \times 10^{-3}$, 0.45 ± 0.08 for $p \le 5 \times 10^{-3}$ and 0.42 ± 0.06 for all p, against $32/75 \approx 0.427$. The ratio ϵ_T/ϵ_S grows from 1.4 at $p = 10^{-2}$ to 2.5 at $p = 10^{-3}$, as a term of order p^2 in ϵ_T alone requires. The exact value of Ref. [15] at $p = 10^{-3}$ is $\epsilon_T = 8.98 \times 10^{-7}$. Its series continues with $(2009039/4500)x^3$ in $x = p/(1-p)$, so at this p the leading term is less than half of ϵ_T.

Flags at distance three. We now show that simple flags cannot repair the distance-three circuit. A flag on qubit q over the window $[t_1, t_2)$ of operations of the check is an extra qubit prepared in $|0\rangle$. It is the target of a CNOT from q before operation t_1 and of another before operation t_2, and it is measured in the Z basis, keeping only the outcome $+1$. It is a detector when no operation in the window changes the Z value of q, that is, when no CNOT targets q and q is neither measured nor reset. It then detects a fault F exactly when the X part of the Pauli frame of F on q differs at t_1 and t_2. We write $\chi_K(F) \in \{0, 1\}$ for this, for each of the 5590 such candidates K. A pair (F_1, F_2) is detected

TABLE SIV. Logical error per accepted run of the circuit of Proposition 5 under uniform depolarizing noise, from the state-vector simulations of Ref. [22] with T and with S gates (one standard deviation), and the acceptance probability with T gates.

p	ϵ_T	ϵ_S	acceptance	$(\epsilon_T - \epsilon_S)/p^2$
0.001	$(9.7 \pm 0.5) \times 10^{-7}$	$(3.9 \pm 0.3) \times 10^{-7}$	0.69	0.58 ± 0.06
0.002	$(5.86 \pm 0.15) \times 10^{-6}$	$(2.87 \pm 0.11) \times 10^{-6}$	0.47	0.75 ± 0.05
0.003	$(1.81 \pm 0.06) \times 10^{-5}$	$(1.03 \pm 0.04) \times 10^{-5}$	0.33	0.88 ± 0.08
0.005	$(8.75 \pm 0.28) \times 10^{-5}$	$(5.81 \pm 0.18) \times 10^{-5}$	0.15	1.18 ± 0.13
0.007	$(2.60 \pm 0.08) \times 10^{-4}$	$(1.90 \pm 0.06) \times 10^{-4}$	0.074	1.42 ± 0.21
0.010	$(9.1 \pm 0.3) \times 10^{-4}$	$(6.7 \pm 0.2) \times 10^{-4}$	0.025	2.4 ± 0.4

by K when $\chi_K(F_1) \neq \chi_K(F_2)$. The fewest flags that detect every harmful pair therefore solve

$$\min_x \sum_K x_K \quad \text{s.t.} \quad \sum_{K:\,\chi_K(F_1)\neq\chi_K(F_2)} x_K \geq 1 \text{ for every harmful pair } (F_1, F_2), \qquad x_K \in \{0, 1\}.$$

The program is infeasible. In two of the pairs the X errors arise on targets of CNOTs, and no candidate detects either fault. Covering the other 46 pairs takes eight flags. Since Chan *et al.* restore fault distance three with two flag qubits and four additional layers of CNOTs [16], flags outside this family suffice. This model ignores the faults that the flags themselves would add. The infeasibility does not depend on them. The script `scripts/cultivation_gsj.py` carries out all computations of this section and writes `results/cultivation/gsj_d3.json`.

The distance-five circuit. The $\delta = 5$ circuit of Ref. [1] has two properties that only appear with real T gates. The first concerns the pattern of T and $T^\dagger$ gates in its checks. On the 19-qubit color code, the transversal check $\bigotimes_q U_q X U_q^\dagger$, with $U_q \in \{T, T^\dagger\}$, is logical only if every face f contains $|f|/2 \bmod 2$ $T^\dagger$ gates [16]. Among such patterns, the parity of the total number of $T^\dagger$ gates decides whether the check stabilizes the injected state $T^\dagger|+\rangle$ or its conjugate. With the wrong parity the check projects the state instead of detecting errors. Half of the runs are then lost, Definition 10 fails and the circuit has fault distance two. The same holds at $\delta = 3$, where transversal T acts on the $[[7, 1, 3]]$ code as the logical $T^\dagger$. The second property concerns the growth from $\delta = 3$ to $\delta = 5$. Its random outcomes leave X-type Pauli frames, and with real T gates these must be undone before the $\delta = 5$ T layers, as in Ref. [11]. We analyze the corresponding branch of maximal weight. With these choices, the full protocol has fault distance three, with leading term $1.5\,p^3$ under uniform noise (Table SVIII). This extends the analysis of the final double check in Refs. [15, 16]. Its logical error agrees with the GPU simulation of Ref. [11] (Table SV).

Sampled logical errors. Next we sample the logical error of the published circuits. We use the published `inject[unitary]+cultivate` circuits, whose parameter d_1 sets the color-code distance δ. At $\delta = 3$ the circuit has 15 qubits and one double check. At $\delta = 5$ it has 42 qubits and double checks at distance three and five, with growth in between. We keep their uniform noise, which includes idling, and read their S gates as real T gates. For the SI1000 model [24] we keep the idle sets of the circuits. We set the two-qubit gate noise to p, single-qubit gates (including T) and idling to $p/10$, resets to $2p$, measurements to $5p$, and idling during measurement or reset layers to $2p$. At $\delta = 5$ the T/$T^\dagger$ pattern on the 19 data qubits places $T^\dagger$ on 18 qubits and T on the qubit shared by the three hexagons in the first layer, and the reverse in the second. The complementary pattern satisfies the same face conditions but checks the conjugate state. The random growth outcomes give several branches. The default reference branch has maximal weight, that is, it has no X frame before the $\delta = 5$ T layers. At $\delta = 3$ under uniform noise, the sampled logical errors agree with the state-vector data of Table SIV. For instance, we find $8.94(33) \times 10^{-7}$ at $p = 10^{-3}$, against $(9.7 \pm 0.5) \times 10^{-7}$. Table SV lists the other cases.

Malignant configurations. Here we describe the malignant configurations of the two published circuits. They involve hooks: X faults on a qubit of the fold tree that the unfold spreads to several data qubits (Sec. S10). At $\delta = 3$ there are nine malignant fault classes, of weight two. Each consists of an X or Y fault on a fold qubit and an X or Y fault on a data qubit at the final T layer. Immediately before that layer they leave X on two or three data qubits. At $\delta = 5$ there are nine malignant classes, of weight three. Each consists of two hooks and one data fault, and they leave X on four or five data qubits. Their acceptance is 2^{-4}. Their bad weight is 2^{-5} when the data fault completes X on four qubits, as Lemma 4 predicts, and 2^{-4} when X already covers a whole logical support before the layer, with Z on an odd number of its qubits, the second case discussed after the proof of the lemma.

Flags at distance five. The lemma also limits what flags can repair. The completing fault can occur anywhere after the unfold, including in any layer inserted there. Flags must therefore catch the hooks themselves. A hook on

TABLE SV. Logical error per accepted output of the circuits of Ref. [1], with real T gates and with the S proxy. Real T and proxy have the same acceptance. At $\delta = 5$ the proxy error is of order p^5. The last column is the GPU simulation of Ref. [11], whose circuit is a modified version of the same protocol.

circuit	noise, p	acceptance	real T	proxy	Ref. [11]
$\delta = 3$	SI1000, 10^{-3}	0.723	$8.00(21) \times 10^{-7}$	$2.84(22) \times 10^{-7}$	
	SI1000, 5×10^{-4}	0.850	$1.52(4) \times 10^{-7}$	$3.35(35) \times 10^{-8}$	
$\delta = 5$	uniform, 2×10^{-3}	0.021	$3.6(7) \times 10^{-8}$		3.4×10^{-8}
	uniform, 10^{-3}	0.144	$3.26(21) \times 10^{-9}$		4.6×10^{-9}
	uniform, 5×10^{-4}	0.379	$2.72(12) \times 10^{-10}$		1.6×10^{-10}
	SI1000, 10^{-3}	0.155	$2.2(6) \times 10^{-9}$		
	SI1000, 5×10^{-4}	0.394	$1.09(8) \times 10^{-10}$		

a fold qubit that is a CNOT target is detected only by a two-qubit parity $Z_a Z_b$. No single-qubit flag measures such a parity without disturbing the check, and such placements exist in the $\delta = 5$ double check:

Proposition 17 (computer-assisted)**.** *No set of non-disturbing single-qubit Z- or X-type flags whose two couplings lie in the double check before the unpairing layer raises the real-T fault distance of the $\delta = 5$ double check of Ref. [1] above three. Two two-qubit flags raise it to four.*

With the two flags, the logical error falls by 2.2 and 3.5 times at $p = 10^{-3}$ and 5×10^{-4} in the SI1000 model [24], at a cost of 2 to 5% in acceptance (see below). The hand-designed flags of Ref. [16] also reach four in the full protocol, with 13 two-qubit flags, and five with a fourth stabilizer round [15, 16]. Without that round, fault distance five requires flagging data errors that enter the check before its first T layer. The flag sets that our integer programs find for this grow to tens of flags and layers.

Flag design. We now describe the flags and the program behind Proposition 17. A Z-type flag on qubit q couples $\mathrm{CX}_{q,f}$ at two layer boundaries and measures the flag qubit f in the Z basis. It flips when the X part of the Pauli frame on q differs at the two boundaries. It is non-disturbing when Z_q at the two boundaries has a deterministic product in the skeleton, which we test by a tableau simulation. X-type flags are defined dually, and two-qubit flags couple to $Z_a Z_b$. The integer program chooses flags that flip for every physical representative of every malignant configuration. Constraints that involve faults of a flag, or of an inserted layer, are imposed only when that flag or layer is chosen. Every design is then re-certified exactly. Consider single-qubit flags with both couplings between the first T layer and the unpairing layer of the $\delta = 5$ double check. Some representatives of the malignant configurations flip none of these candidate flags. They remain malignant in every flagged circuit, which proves Proposition 17. With two-qubit flags, the program finds two flags, $Z_3 Z_7$ and $Z_{32} Z_{33}$ (qubit labels of the published circuit), coupled over the whole fold and unfold. They give fault distance four. At uniform $p = 5 \times 10^{-4}$, the flagged circuit has logical error $1.07(15) \times 10^{-10}$ against $2.72(12) \times 10^{-10}$ for the unflagged circuit, with acceptance 0.345 against 0.379. In SI1000 at $p = 10^{-3}$ and 5×10^{-4}, it has $1.00(28) \times 10^{-9}$ and $3.1(6) \times 10^{-11}$ against $2.17(56) \times 10^{-9}$ and $1.09(8) \times 10^{-10}$, at acceptance 0.148 and 0.385 instead of 0.155 and 0.394. The remaining malignant configurations, of weight four, begin with X faults on two data qubits between the last stabilizer round and the first T layer. No flag inside the check can see them.

S10. COHERENT COMPLETION AND GHZ FOLDS

This section turns Lemma 4 into a bound on the fault distance, explains what the bound asks of a check, and describes the GHZ-fold checks that we built at distance three, five and seven. Throughout, a circuit is a cultivation circuit on a color code of distance δ, as in Lemma 4.

Corollary 18. *The fault distance of such a circuit is at most $s + \delta - |u|$ for every configuration of s faults that leaves an undetected X^u as in Lemma 4 immediately before the last T layer. In particular, it is at most $1 + h$, where h is the least number of faults that leave such an X^u with $|u| = \delta - 1$.*

A check that keeps fault distance δ with real T gates must therefore be fault tolerant *relative to the code*. That is, no s undetected faults may place X on more than s qubits of a minimum-weight logical support before the last T layer. The proxy imposes no such condition, because the S layer turns these errors into Pauli errors with a detectable Z part. The condition is much weaker than ordinary fault tolerance of the check ancilla. At $\delta = 3$ every pair of data

qubits lies in a minimum-weight logical support. At $\delta = 5$ only 4.6% of the weight-four patterns do, and at $\delta = 7$ only 0.04% of the weight-six ones (Table SVII). Lower bounds on optimal fault-tolerant cat preparations [41] therefore do not apply to real-T checks.

The double checks of Ref. [1] measure the transversal check with a GHZ fold over data and partner qubits. A single X fault on a fold node spreads, after unfolding, to the data of a whole subtree. We call this a hook. For the circuits of Ref. [1], $h = 1$ at $\delta = 3$ and $h = 2$ at $\delta = 5$. Every malignant configuration has the form of Lemma 4 or of its second neighboring case, with the predicted acceptance $2^{-(\delta-1)}$ and bad weight (Table SVIII). The bound of Corollary 18 is therefore attained.

Ref. [1] gives circuits up to $\delta = 5$, and how much fault distance is lost at larger distance was open [16]. We therefore built GHZ-fold checks of the same structure on the $[[7, 1, 3]]$, $[[19, 1, 5]]$ and $[[37, 1, 7]]$ color codes, described below. For every fold tree we tried (three trees at $\delta = 3$ and $\delta = 5$, six at $\delta = 7$, of depth one to seven) they have real-T fault distance two, three and four, against proxy fault distance three, five and at least five [Proposition 5(ii)]. The real-T fault distance of the fold thus grows with the code distance, but at half its rate (Fig. 3). Its failures are partly suppressed: at $\delta = 7$, every malignant configuration of four faults is accepted with probability between 2^{-10} and 2^{-4}, and the leading term of the probability of an accepted wrong output ranges from $0.011\,p^4$ to $15\,p^4$ over the six trees. The choice of tree costs nothing and changes this term by three orders of magnitude, but the loss of distance remains.

GHZ-fold checks at $\delta = 3$, 5 and 7. We now give the construction in detail. The circuits of Proposition 5 use the same surroundings as the cat checks of Sec. S11, an ideally encoded input and flagged stabilizer rounds, and their one check round is built like the double check of Ref. [1]. It consists of n partners prepared in $|+\rangle$, the $\mathrm{U}^\dagger$ layer, and CX from each partner to its data qubit. Then comes a fold CX(parent, child) over a spanning tree, leaves first, whose nodes are one representative (data qubit or partner) of each pair. The root is measured in the X basis and reset. Finally come the fold in reverse, the pairing CNOTs again, the U layer and X measurements of the partners. The root and the product of the partners read the check twice. The trees are spanning trees of the graph of data qubits that share a face. They are grown breadth-first from a central qubit or in random order. They are either unrestricted (depth one to three) or have at most two children per node (depth five to seven, as for the nearest-neighbor fold of Ref. [1]), and their representatives are either data qubits or chosen at random. The detectors are all record parities that are deterministic in the noiseless proxy. The injected state is $\mathrm{T}^\dagger|+\rangle$ at $\delta = 3$ and 7 and $\mathrm{T}|+\rangle$ at $\delta = 5$, the one for which Definition 10 holds. At $\delta = 3$ the three trees have 6, 12 and 12 malignant pairs, with leading terms $0.089p^2$, $0.57p^2$ and $1.0p^2$. At $\delta = 5$ they have 68, 85 and 3 malignant triples, with $0.43p^3$, $0.69p^3$ and $0.076p^3$. At $\delta = 3$ and 5 these fault distances are those of the circuits of Ref. [1]. At $\delta = 7$ ($t = 75$ T gates, affine dimension $\kappa = 20$ of the Choi state, Sec. S8) no tree has a harmful configuration of three faults. At weight four, the six trees have 15, 144, 80, 523, 362 and 352 distinct harmful errors, each realized by four fault classes whose exact detector patterns cancel. Each certificate takes 8 to 17 minutes. The proxy distances follow from the stim detector error model of the proxy circuit, with the same search.

Leading terms and acceptance at $\delta = 7$. The leading terms of Table SVIII at $\delta = 7$ come from the exact enumeration of all malignant configurations of four faults. The six fold trees of Proposition 5 are called central, central binary, binary 1, binary 2, binary 3 and random 1. They have 15, 256, 111, 957, 501 and 517 malignant configurations, with total weights 0.011, 0.24, 0.062, 15, 4.5 and 5.4 times p^4. The unverified cat has 120, with $0.093\,p^4$. On the flagged double check at $\delta = 5$, the same enumeration gives 6,824 configurations and $1.16 \times 10^3 p^4$, as in Table SVIII. The acceptances of Table SVI come from sampling the detector error model (5×10^6 samples), with the exact acceptance of each sampled class.

TABLE SVI. Acceptance of the checks of Table SVIII, built in our surroundings (one check round between flagged stabilizer rounds), under uniform noise. The GHZ fold uses the central binary tree.

check	$\delta = 5$, 10^{-3}	$\delta = 5$, 5×10^{-4}	$\delta = 7$, 10^{-3}	$\delta = 7$, 5×10^{-4}
GHZ fold	0.050	0.222	0.0005	0.020
cat, not verified	0.226	0.474	0.015	0.120
cat, verified	0.110	0.330	0.0009	0.029

TABLE SVII. Fraction of the X patterns of weight w on the data of the triangular color codes that lie inside the support of a minimum-weight logical X operator. By Corollary 18, in a check of full real-T fault distance such a pattern may not be left by fewer undetected faults than its weight.

code (minimum-weight X logicals)	$w=2$	3	4	5	6
$[[7,1,3]]$ (7)	1				
$[[19,1,5]]$ (36)	0.895	0.291	0.046		
$[[37,1,7]]$ (140)	0.811	0.278	0.052	0.0060	0.0004

S11. VERIFIED CATS AND NEAREST-NEIGHBOR CHECKS

This section gives the details behind Proposition 6. We first describe the verified-cat checks with all-to-all connectivity, then their nearest-neighbor versions on a square grid and what they cost, and finally a complete nearest-neighbor protocol and the circuits themselves.

Verified cats. Each Shor-style check round [23] first applies the $\mathrm{U}^\dagger$ layer, where U is the transversal layer of T and $\mathrm{T}^\dagger$ gates on the data. It then prepares a GHZ state, or cat state, on fresh qubits c_q and verifies it by measuring all adjacent parities $Z_{c_q}Z_{c_{q+1}}$ before it touches the data. Next it couples the cat once to each data qubit by $\mathrm{CX}_{c_q,q}$ and measures the cat in the X basis. The parity of these outcomes is the check outcome. Finally the round applies the U layer. A fault in the cat preparation that spreads X over several cat qubits is caught by the verification, and every other fault reaches at most one data qubit. Flagged stabilizer rounds [39] surround each check round.

On the $[[19,1,5]]$ color code, the verified-cat check has fault distance exactly five with real T gates, both with one check round and with two. With one round, the probability of an accepted wrong output is $3452\,p^5 + O(p^6)$ under uniform circuit noise. On the $[[37,1,7]]$ code, one round has fault distance at least five. Without verification, the same circuits have fault distance three at $\delta = 5$ and four at $\delta = 7$. On the $[[7,1,3]]$ code, the verified-cat check keeps fault distance three, where the double check of Ref. [1] has two. The certificates exclude every harmful configuration of at most four faults, and the upper bounds come from Proposition 9. The S proxy does not distinguish the verified and the unverified circuits. With real T gates, their fault distances differ by two at $\delta = 5$ (Table SVIII). All the unverified checks we certified, the GHZ folds at $\delta = 3$, 5 and 7 and the unverified cat at $\delta = 5$ and 7, have fault distance $(\delta+1)/2$.

Verification has two prices. The first is acceptance, which is about half that of the unverified cat under uniform noise. The second is connectivity, since the cat check as written is not nearest-neighbor on a square grid. The nearest-neighbor checks below remove both costs. Acceptance also limits how far the distance can be raised. In our circuits, under uniform noise at $p = 10^{-3}$, going from $\delta = 5$ to $\delta = 7$ lowers the acceptance from 5% to 0.05% for the GHZ fold, from 23% to 1.5% for the unverified cat and from 11% to 0.09% for the verified cat. At $p = 5\times10^{-4}$ the values at $\delta = 7$ are 2.0%, 12% and 2.9% (Table SVI). Our circuits are not optimized for acceptance. For comparison, the published $\delta = 5$ protocol accepts 14% at $p = 10^{-3}$. At $\delta = 3$, fault distance three is also expensive in its coefficient. The verified cat has 2,405 malignant triples and a leading term $4.2\times10^3 p^3$, which is larger than the $0.43p^2$ of the double check of Ref. [1] for $p \gtrsim 10^{-4}$.

We now move the cat check onto a square grid. There a cat state can be verified with the same economy if it is grown along a path of grid sites from the middle outward. A single X fault on a path qubit then either stays on that qubit or spreads to the part of the path beyond it, which contains one of the two ends. The parity Z_aZ_b of the two ends a and b, measured through an ancilla next to both, therefore detects every spread. Each data qubit sits next to its own path site and interacts with the check once (Fig. 4). After the coupling we un-grow the cat: we apply the CNOTs that grew it in reverse order, so that the cat collapses onto its root, the site where the growth started. The check outcome is then a single X measurement of the root instead of the parity of the whole path. The other path qubits return to $|0\rangle$, and their measurements give further detectors. Faults in this step touch no data qubit. At $\delta = 3$ the path runs around a rectangle and the ancilla closes it. At $\delta = 5$ the path is folded into a U on three rows, with its arms on the rows $(x,0)$ and $(x,2)$. The middle row measures the rungs $Z_{(x,0)}Z_{(x,2)}$, each through one ancilla, and the diagonals $Z_{(x,0)}Z_{(x-1,2)}$, each through two adjacent ancillas. The rungs detect any spread confined to one arm of the U. The diagonals detect the equal spreads on both arms that two faults at mirror positions produce, and without them the ladder has fault distance four. Neither check needs stabilizer rounds: the code is checked only by the final ideal decoder, as in the circuits of Ref. [1]. With two-qubit gates only between neighbors of a square grid, the ring check on the $[[7,1,3]]$ code (17 qubits) and the ladder check on the $[[19,1,5]]$ code (46 qubits) have real-T fault distance three and five, equal to the code distance. Under uniform noise, the probability of an accepted wrong

TABLE SVIII. Real-T fault distance of cultivation checks, certified exactly with Theorems 1 and 2, and the leading term of the probability of an accepted wrong output under uniform circuit noise of strength p. The proxy distance is that of the same circuit with S in place of T. Malignant: the number of malignant configurations. Those of the first two rows are accepted with probability $2^{-(\delta-1)}$ relative to the fault-free run. The first three rows are the protocol of Ref. [1], the third with our two flags. The $\delta = 7$ GHZ-fold row is our construction of Proposition 5, and the cat rows have one check round (two where stated). Both sit between flagged stabilizer rounds and start from an ideally encoded input. For the $\delta = 7$ GHZ fold, the ranges are over the six trees. The nearest-neighbor (NN) checks of Proposition 6 have one check round, an un-grown cat and no stabilizer rounds. The NN protocol prepares the state itself, with every operation noisy. The last row is the two-check circuit of Ref. [2], with its injection and error-correction cycles, under SI1000 noise (Proposition 19). Fault distances are at most the code distance (Proposition 9).

check	δ	proxy	real T	malignant	leading term
GHZ fold [1]	3	3	2	9	$0.43\,p^2$
GHZ fold [1]	5	5	3	9	$1.5\,p^3$
GHZ fold, 2 flags	5	5	4	6824	$1.2 \times 10^3 p^4$
GHZ fold, 6 trees	7	≥ 5	**4**	15 to 957	$(0.011 \text{ to } 15)\,p^4$
cat, verified	3	3	**3**	2405	$4.2 \times 10^3 p^3$
cat, not verified	5	5	3	99	$0.54\,p^3$
cat, verified	5	5	**5**	754,515	$3.5 \times 10^3 p^5$
cat, verified, 2 rounds	5	5	**5**		
cat, not verified	7	≥ 5	4	120	$0.093\,p^4$
cat, verified	7	≥ 5	**5** to **7**		
NN cat, ring	3	3	**3**	574	$3.6\,p^3$
NN cat, ladder	5	5	**5**	85,464	$12\,p^5$
NN cat, ladder	7	≥ 5	**6** to **7**		
NN protocol	3	3	**3**	10,573	$2.8 \times 10^3 p^3$
Google experiment	3	3	2	20	$2.2\,p^2$

output is $3.6\,p^3$ and $12\,p^5$. At $\delta = 3$ this is the same as with the S proxy. Under SI1000 noise it is $0.056\,p^3 + 30\,p^4$ at $\delta = 3$, and its leading term at $\delta = 5$ is $0.012\,p^5$. At $p = 10^{-3}$ in SI1000 the checks accept 88% and 39% of runs. The ladder on the $[[37, 1, 7]]$ code (91 qubits) has fault distance at least six. Without un-growing, every X measurement of the cat flips the check outcome when it fails, which happens with probability $5p$ in SI1000, and these measurements dominate the failures. The leading terms are then $6.9\,p^3$, with the end parity of the ring measured twice, and $43\,p^5$, at acceptance 89% and 55%. At each distance these versions have the highest acceptance and the smallest spacetime volume in Table SIX.

The grid checks also compare well with our other checks. At $\delta = 3$ the ring beats every GHZ fold at every $p < 10^{-2}$, with half the qubits and twice the acceptance of the best one. At $\delta = 5$, fault distance five costs less than fault distance three: the ladder accepts four to six times more often than the unverified cat and 20 to 28 times more often than the best fold, and fails three orders of magnitude less often. Against the verified cat of Proposition 6 it uses a third of the qubits and at most two thirds of the depth, accepts 13 to 18 times more often, and has a leading term 300 times smaller under uniform noise. Most of these differences come from the stabilizer rounds, which the ladder does not need. At $\delta = 7$ the ladder accepts 7% to 21% of runs, where our checks with stabilizer rounds accept 3×10^{-6} to 10^{-3}. In SI1000 the minimal failures of the un-grown checks consist of faults of strength $p/10$, so the next order matters at $p = 10^{-3}$, and at $\delta = 3$ it adds the $30\,p^4$ above.

A complete protocol. The checks above start from an ideally encoded state. On the grid, the state can also be prepared with every operation noisy. A $T^\dagger$ gate is applied to one data qubit in $|+\rangle$. The CNOTs of a non-fault-tolerant encoder are then relayed along the path: the control is copied onto its path site, spread along the segment that covers its targets, applied, and unspread. The encoder has hooks, single faults that leave X or Z on several data qubits. All six stabilizers are therefore measured, each with its own verified path cat, before three checks. This 17-qubit protocol has real-T fault distance three, equal to its proxy distance. Without the stabilizer measurements, or with only the Z stabilizers, it has fault distance two, because one more fault completes a Z hook of weight two into a logical Z that no check sees. The protocol pays for the stabilizer measurements. It has 107 layers, many of them measurement layers, during which every data qubit idles at $2p$ in SI1000. Its leading term is therefore $7.8 \times 10^3 p^3$ at 11% acceptance, which is ahead of the experiment's circuit for $p \lesssim 3 \times 10^{-4}$ (Table SIX). Another route prepares $|0\rangle_L$ by measuring

TABLE SIX. Cost of cultivation checks under SI1000 noise at $p = 10^{-3}$: qubits, circuit layers, acceptance relative to the fault-free run, real-T fault distance (FD), leading terms of the probability of an accepted wrong output, the resulting logical error per accepted state ϵ, and the expected spacetime volume per accepted state, qubits × layers / acceptance. Rows marked * prepare the state themselves: by injection, and growth at $\delta = 5$, for Refs. [1, 2], whose ϵ is sampled, and by relayed injection for the NN protocol. The other rows start from an ideally encoded state. NN: nearest-neighbor on a square grid. GHZ fold: the tree with the smallest leading term. At fault distance five, ϵ is the leading term only, and it is omitted where the next order dominates.

check	δ	qubits	layers	acc.	FD	leading terms	ϵ	volume
Ref. [1]*	3	15	39	0.72	2		8.0×10^{-7}	8.1×10^{2}
Google*, NN	3	14	156	0.52	2	$2.2\,p^2$	3.1×10^{-6}	4.2×10^{3}
GHZ fold	3	38	46	0.43	2	$0.049\,p^2$	4.9×10^{-8}	4.1×10^{3}
cat, verified	3	69	34	0.22	3	$1.7 \times 10^{4} p^3$	1.7×10^{-5}	1.1×10^{4}
NN ring, not un-grown	3	17	14	0.89	**3**	$6.9\,p^3$	6.9×10^{-9}	$\mathbf{2.7 \times 10^{2}}$
NN ring	3	17	15	0.88	**3**	$0.056\,p^3 + 30\,p^4$	$\mathbf{8.6 \times 10^{-11}}$	2.9×10^{2}
NN protocol*	3	17	107	0.11	**3**	$7.8 \times 10^{3} p^3$	7.8×10^{-6}	1.7×10^{4}
Ref. [1]*	5	42	94	0.16	3		2.2×10^{-9}	2.5×10^{4}
GHZ fold	5	110	74	0.020	3	$0.052\,p^3$	5.2×10^{-11}	4.1×10^{5}
cat, not verified	5	110	36	0.093	3	$0.46\,p^3$	4.6×10^{-10}	4.2×10^{4}
cat, verified	5	128	51	0.030	5	$8.3 \times 10^{3} p^5$	8.3×10^{-12}	2.1×10^{5}
NN ladder, not un-grown	5	46	24	0.55	**5**	$43\,p^5$	4.3×10^{-14}	$\mathbf{2.0 \times 10^{3}}$
NN ladder	5	46	34	0.39	**5**	$0.012\,p^5$		4.0×10^{3}
GHZ fold	7	218	114	3×10^{-6}	4	$0.0087\,p^4$	8.7×10^{-15}	8×10^{9}
cat, not verified	7	218	51	9.1×10^{-4}	4	$0.086\,p^4$	8.6×10^{-14}	1.2×10^{7}
cat, verified	7	254	88	1×10^{-5}	≥ 5			2×10^{9}
NN ladder, not un-grown	7	91	33	0.21	≥ 5			$\mathbf{1.5 \times 10^{4}}$
NN ladder	7	91	52	0.068	≥ 6			7.0×10^{4}

the X stabilizers and lets the first check project onto the T state, which needs no T gate. It also reaches fault distance three, at a higher cost. At $\delta = 3$ on a grid, fault distance is thus no longer the obstacle. The obstacle is the cost of measuring stabilizers around the checks.

Verified-cat checks. Here we give the circuits of Proposition 6 and the numbers behind their certificates. At $\delta = 5$ and $\delta = 7$ the circuits start from an ideally encoded state, made from the injected state on one qubit and a CNOT encoder. A flagged stabilizer round follows, then one check round (or two, separated by a flagged stabilizer round), and then a last flagged stabilizer round. Uniform noise of strength p acts after every gate, on resets and measurements and on idle live qubits, and the final readout is noiseless. The cat on n qubits is prepared by a CNOT tree from one qubit in $|+\rangle$ and verified by the $n-1$ parities $Z_{c_q} Z_{c_{q+1}}$, each with its own ancilla. The check outcome is the parity of the n X measurements of the cat. The injected state, $\mathrm{T}|+\rangle$ at $\delta = 5$ and $\mathrm{T}^\dagger|+\rangle$ at $\delta = 7$, is matched to the global parity of the $\mathrm{T}/\mathrm{T}^\dagger$ pattern. At $\delta = 5$, the certificate of fault distance five for one round evaluates exactly the 56,319 distinct observable patterns of undetected configurations of four faults and finds none harmful. The enumeration at weight five finds 754,515 malignant configurations, with total weight 3.452×10^{-12} at $p = 10^{-3}$. Without verification, 99 malignant configurations of weight three have total weight 5.4×10^{-10}. For two rounds the weight-four certificate evaluates 3,220,729 distinct errors. At $\delta = 7$ ($t = 75$ T gates, affine dimension $\kappa = 20$), the weight-four certificate of the verified cat evaluates 464,480 distinct errors and finds none harmful. The unverified cat has no harmful configuration of weight three. At weight four it has 38 distinct harmful errors. Each is realized by four fault classes whose exact detector patterns cancel, and each is accepted with probability 2^{-5}, 2^{-6} or 2^{-7} relative to the fault-free run. At uniform $p = 10^{-3}$ the acceptance at $\delta = 5$ is 0.11 with verification and 0.23 without. In SI1000 at $p = 5 \times 10^{-4}$ it is 0.17 and 0.30. The $\delta = 3$ verified-cat circuit applies T to all seven data qubits in each layer. It injects $\mathrm{T}^\dagger|+\rangle$ on one qubit before a non-fault-tolerant encoder, and it has two check rounds and three flagged stabilizer rounds. Its fault distance with real T gates is three, equal to its proxy distance, with 2,405 malignant configurations of weight three. With $\mathrm{T}|+\rangle$ injected instead, its checks project. Definition 10 then fails, with four accepted Z classes, and the fault distance is two.

Nearest-neighbor checks. Here we give the circuits of Proposition 6. They start from the same ideally encoded state as the cat checks above. They apply the $\mathrm{U}^\dagger$ layer, prepare the cat on the path, verify it, and couple each data

qubit once to its path site by a CNOT. They then un-grow the cat, measure the root in the X basis and the other path qubits in the Z basis, and apply the U layer. The final readout is noiseless, and there are no stabilizer rounds. Every two-qubit gate of the check acts on grid neighbors, and the construction asserts this. A CNOT counts as one two-qubit gate. On CZ hardware it carries two Hadamards, each with noise $p/10$ in SI1000. The path is grown from its middle site, one CNOT per arm per layer. In the ring ($\delta = 3$) the path has nine sites, seven next to data and two corner bridges, and the ancilla next to both ends measures $Z_a Z_b$. One round takes 15 layers. The $30\,p^4$ term of Proposition 6 is the sum over all 9,340 malignant configurations of four faults. Without un-growing, with the cat read out by nine X measurements, the leading term is $16\,p^3$ in SI1000 ($30\,p^3$ with two check rounds), and $6.9\,p^3$ when the end parity is measured twice. In the ladder ($\delta = 5$) the 19 path sites are the top row $(x, 0)$, $x = 0, \ldots, 7$, the turn $(8, 0), (8, 1), (8, 2)$ and the bottom row $(x, 2)$, $x = 7, \ldots, 0$. Each is next to one data qubit, and the root is at $(8, 1)$. The eight middle sites $(x, 1)$ measure the eight rungs and, in pairs, the seven diagonals, with the odd and the even diagonals in two parallel layers. Each two-ancilla gadget uncomputes its second ancilla. One round takes 34 layers (24 without un-growing). The weight-four certificate evaluates 15,775 distinct errors in 7 s. The enumeration at weight five finds 85,464 malignant configurations. Without un-growing it finds 92,216, and in that case a second check round also keeps fault distance five, with 273,827 distinct errors at weight four. Without the diagonals the ladder has fault distance four, with nine malignant configurations of weight four. The $\delta = 7$ ladder has 37 path sites ($17 + 3 + 17$), 17 middle ancillas and 91 qubits, and it injects $\mathrm{T}^\dagger|+\rangle$. Its weight-four certificate evaluates 195,829 distinct errors in under three minutes. With the un-grown cat, we enumerated all configurations of five faults whose detector patterns cancel and found no harmful one, in a quarter of an hour on 14 cores.

The complete nearest-neighbor protocol. The complete protocol of the main text uses the ring layout with every operation noisy. The data are reset. $\mathrm{T}^\dagger$ is applied to the carrier, and H to the carrier and to the pivots of the encoder. Each data-data CNOT fan-out of the encoder is relayed. The path segment that covers the control and its targets is reset, and the control is copied onto its path site. The copy is spread along the segment and applied to the targets from their own path sites. It is then unspread and uncopied, and the segment is measured, with outcome deterministically zero. The three Z and the three X stabilizers are then measured, each with a verified path cat coupled to the four data qubits of the face. For the Z stabilizers the CNOTs are conjugated with Hadamards on the data. Three checks follow, with the cat read out by X measurements. The circuit has 17 qubits and 107 layers, and every two-qubit gate acts between grid neighbors. Its real-T and proxy fault distances are both three. It has 10,573 malignant configurations of weight three, with leading term $2.8 \times 10^3 p^3$ under uniform noise and $7.8 \times 10^3 p^3$ in SI1000, at acceptance 0.11 in SI1000 at $p = 10^{-3}$. With two checks the leading term in SI1000 is $4.8 \times 10^4 p^3$, and with un-grown cats the deeper circuit has $6.1 \times 10^4 p^3$. If the Z stabilizers are measured instead by folding their parity along the path, weight-two Z hooks of the encoder go undetected and the fault distance is two. The alternative prepares $|0\rangle_L$ by measuring the X stabilizers with the path cat and lets the first check project. There, errors before the first check cause two problems. A Z error imitates a Z_L frame, and an X error passes all T layers coherently. The X and Z stabilizers must therefore be measured again after the first check. The schedule X stabilizers, check, Z and X stabilizers, check, check has fault distance three and leading term $7.1 \times 10^4 p^3$ in SI1000, at acceptance 0.05.

S12. A CIRCUIT RUN ON HARDWARE

As an example of a circuit that has been run on hardware, this section analyzes the cultivation circuit of the experiment of Ref. [2]. The experiment ran a $\delta = 3$ color code on a square grid with nearest-neighbor CZ gates. It used injection, GHZ-fold checks with time-reversed unfolding, and error-correction cycles between checks, and it projected its logical error with the S proxy. We converted the published circuits [2] into real-T circuits. Each physical $R_y(\pm\pi/4)$ rotation is written as T or $\mathrm{T}^\dagger$ conjugated by the circuit's own H_{YZ} gates, and the injection sign is chosen so that the fault-free check accepts with certainty. We added SI1000 noise, the model of the authors' simulations, and kept the virtual T gates noiseless.

Proposition 19 (computer-assisted)**.** *After one, two or three checks, the circuit of Ref. [2] has real-T fault distance two, while its proxy fault distance is three. After two checks, each of its 20 malignant pairs contains a fault in the unfolding of the last check, and an ideal injection leaves the same 20 pairs.*

The experiment is thus a GHZ fold with fault distance $(\delta + 1)/2$, as Proposition 5 predicts. At the noise strength of the experiment, $p = 2.3 \times 10^{-3}$, the two-check circuit accepts 22% of runs. Its logical error is 2.4×10^{-5} with real T gates, against 6.9×10^{-6} with the proxy. The ratio grows from 3.1 at $p = 3 \times 10^{-3}$ to 4.5 at 1.5×10^{-3} and 6.1 at 10^{-3}, because the leading term is $2.2\,p^2$ instead of order p^3 (Fig. S2). Projections of the experiment to lower noise along the p^3 slope of the proxy are therefore optimistic, by a factor that grows as $1/p$. On the same kind of grid

and from the same ideal input, the two checks of the experiment have leading term $0.44\,p^2$, while one ring check of Proposition 6 has $0.056\,p^3 + 30\,p^4$. At $p = 2.3 \times 10^{-3}$ the ring thus fails with about 10^{-3} times the probability, with 17 qubits instead of 14.

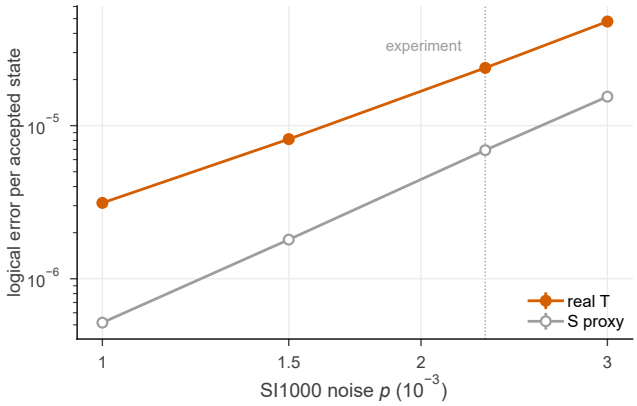


FIG. S2. Logical error per accepted state of the cultivation circuit of Ref. [2] (two checks with error-correction cycles) under SI1000 noise, with real T gates (vermilion) and with S gates (grey), at equal acceptance. The experiment ran at $p \approx 2.3 \times 10^{-3}$ (dotted).

Decoding the published circuits. Here we describe how we decode and simulate the published circuits. Ref. [2] publishes its circuits as stim files. In these files every non-Clifford rotation is an identity tagged `T_gate`, `T_dagger_gate` or `injection`, and the physical Hadamards that turn the Y rotations into virtual Z rotations are tagged `conjugate`. We replace the tagged identities by H_{YZ}, T and $\mathrm{T}^\dagger$. We write the injection $R_y(\pi/4)|0\rangle$ as $\sqrt{Y}^\dagger$, $\mathrm{T}^{\pm 1}$ and H_{YZ}, which prepare the same state, with the sign for which the fault-free circuit accepts with certainty. The other sign halves the acceptance. The circuit of their Fig. 3 has 14 qubits, seven of them data qubits. It consists of injection, three error-correction cycles and three checks, the last of which is the kickback measurement. We cut it after the first, second or third check, or after the following cycle, and read out the six stabilizers and Z_L noiselessly, as for the circuits of Ref. [1]. The detectors are all deterministic record parities, so every run with a detection event is discarded, as in the experiment. SI1000 noise is applied layer by layer, with echo pulses treated as single-qubit gates, and layers that hold only T gates take no time. Every cut has real-T fault distance two with 20 malignant pairs and proxy fault distance three, with leading terms from $0.44\,p^2$ to $2.2\,p^2$. Making the injection and the first cycle noiseless leaves the 20 pairs and their weights unchanged. The single check of their Fig. 2, without cycles, has fault distance two both with real T gates and with the proxy. The logical errors of Fig. S2 come from 1.2×10^9 to 1.1×10^{10} samples per point. Ref. [2] reports a logical error of 1 to 2×10^{-4} at 8% retention for the kickback circuit with its own measurement. That number includes the error of the kickback readout and noise outside the Pauli model. Our numbers are those of the cultivated state before that readout.

S13. OPEN PROBLEMS

We close with five open problems.

Stabilizer measurements on the grid. The nearest-neighbor checks of Proposition 6 keep the full distance from an ideally encoded state, and at distance three the complete protocol of Sec. S11 keeps it with every operation noisy. That protocol spends most of its layers on stabilizer measurements, during which the data idle. A cheaper way to measure the stabilizers around the checks on a square grid, or a preparation that needs fewer of them, would make full-distance cultivation competitive at the noise strengths of current hardware.

Larger circuits. Theorem 2 decides the collapse condition in polynomial time for circuits with deterministic checks, and this is how the circuits of Propositions 5 and 6 were certified. The cost of a certificate now lies in the search over fault configurations whose detector patterns cancel. At $\delta = 7$ the searches at weight four take minutes, and those at weight five a quarter of an hour. The nearest-neighbor ladder at $\delta = 7$ has no harmful configuration of five faults, and a search at weight six would decide whether it keeps the full distance.

Lower bounds for real-T checks. By Corollary 18, a check that keeps the full distance with real T gates must be fault tolerant relative to the code, a condition much weaker than fault tolerance of the cat state itself (Table SVII).

Known lower bounds on fault-tolerant cat preparation [41] therefore do not apply, and we know of no lower bound on the number of two-qubit gates of such a check.

Space-time cost at small distance. The circuit of Proposition 16 minimizes the number of T gates. Two questions remain open. The first is the least number of qubits, two-qubit gates or time steps at fault distance two. The second is how many T gates fault distance three needs, which is between ten [19] and fifteen (Sec. S5). By Theorem 14, the second question is the distillation question of Ref. [19] at order three. Integer programs over the fault map, as in Sec. S5, apply to any family of circuits in which the fault map is linear in the design choices, such as orders of gates and positions of flags.

Rates. Proposition 7 discards the acceptance probability, so bounds on the expected cost per accepted state need a different argument.